tropical: BalancedPartition

2026-10-25T07:11:26

1 Key results

This report covers 220 runs (5 seeds; 0 diverged; 20 never completed, see `registry.jsonl`) under frozen hash `beab4c1c6902`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the parametric path: values away from the metric bounds. Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8967	[0.863, 0.929]	0.0429	5	0
adaptive	pointer	0.01	0.8915	[0.864, 0.918]	0.0359	5	0
adaptive	pointer	0.1	0.885	[0.838, 0.926]	0.057	5	0
adaptive	pointer	1	0.896	[0.862, 0.927]	0.0417	5	0
adaptive	pointer	10	0.8922	[0.866, 0.918]	0.0328	5	0
adaptive	pointer	100	0.8996	[0.866, 0.929]	0.0408	5	0
adaptive	pointer	1000	0.8935	[0.847, 0.933]	0.0555	5	0
adaptive	pointer	1e+04	0.9083	[0.883, 0.932]	0.0317	5	0
adaptive	seq2seq	0	0.9038	[0.898, 0.911]	0.00814	5	0
adaptive	seq2seq	0.01	0.8995	[0.897, 0.902]	0.00296	5	0
adaptive	seq2seq	0.1	0.9056	[0.905, 0.907]	0.00109	5	0
adaptive	seq2seq	1	0.9047	[0.9, 0.909]	0.00602	5	0
adaptive	seq2seq	10	0.9095	[0.9, 0.918]	0.0116	5	0
adaptive	seq2seq	100	0.9087	[0.902, 0.916]	0.00956	5	0
adaptive	seq2seq	1000	0.9173	[0.915, 0.92]	0.00334	5	0
adaptive	seq2seq	1e+04	0.905	[0.894, 0.917]	0.0155	5	0
tropical	pointer	0	0.9032	[0.875, 0.926]	0.0337	5	0
tropical	pointer	0.01	0.9099	[0.9, 0.918]	0.011	4	0
tropical	pointer	0.1	0.9146	[0.898, 0.931]	0.0202	4	0
tropical	pointer	10	0.9092	[0.894, 0.926]	0.02	4	0
tropical	pointer	1000	0.8829	[0.835, 0.91]	0.0417	3	0
tropical	seq2seq	0	0.8978	[0.879, 0.908]	0.0204	5	0
tropical	seq2seq	0.01	0.9072	[0.906, 0.908]	0.00129	5	0
tropical	seq2seq	0.1	0.9131	[0.907, 0.925]	0.0136	5	0
tropical	seq2seq	1	0.9059	[0.904, 0.907]	0.00199	5	0
tropical	seq2seq	10	0.9066	[0.905, 0.908]	0.00219	5	0
tropical	seq2seq	100	0.9194	[0.908, 0.931]	0.0153	5	0
tropical	seq2seq	1000	0.9119	[0.905, 0.925]	0.0142	5	0
tropical	seq2seq	1e+04	0.8779	[0.861, 0.895]	0.0235	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
vanilla	pointer	0	0.9167	[0.904, 0.926]	0.0137	5	0
vanilla	pointer	0.01	0.9225	[0.916, 0.926]	0.00687	5	0
vanilla	pointer	0.1	0.9214	[0.916, 0.926]	0.00645	5	0
vanilla	pointer	1	0.9223	[0.92, 0.925]	0.0035	5	0
vanilla	pointer	10	0.9216	[0.917, 0.926]	0.00646	5	0
vanilla	pointer	100	0.9208	[0.918, 0.923]	0.00337	5	0
vanilla	pointer	1000	0.9225	[0.919, 0.925]	0.00395	5	0
vanilla	pointer	1e+04	0.9141	[0.906, 0.92]	0.00872	5	0
vanilla	seq2seq	0	0.9261	[0.922, 0.93]	0.00543	5	0
vanilla	seq2seq	0.01	0.927	[0.926, 0.928]	0.00143	5	0
vanilla	seq2seq	0.1	0.9283	[0.926, 0.931]	0.00354	5	0
vanilla	seq2seq	1	0.9277	[0.924, 0.931]	0.00433	5	0
vanilla	seq2seq	10	0.93	[0.926, 0.933]	0.00459	5	0
vanilla	seq2seq	100	0.9311	[0.929, 0.933]	0.00209	5	0
vanilla	seq2seq	1000	0.9281	[0.923, 0.932]	0.00637	5	0
vanilla	seq2seq	1e+04	0.9321	[0.928, 0.936]	0.00505	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.005196	[-0.0807, 0.0703]	1.0000
adaptive	pointer	0.1	-0.01174	[-0.0872, 0.0637]	0.9981
adaptive	pointer	1	-7.894×10^{-4}	[-0.0763, 0.0747]	1.0000
adaptive	pointer	10	-0.0045	[-0.08, 0.071]	1.0000
adaptive	pointer	100	0.002846	[-0.0726, 0.0783]	1.0000
adaptive	pointer	1000	-0.003251	[-0.0787, 0.0722]	1.0000
adaptive	pointer	1e+04	0.01154	[-0.064, 0.087]	0.9983
adaptive	seq2seq	0.01	-0.004328	[-0.0193, 0.0107]	0.9444
adaptive	seq2seq	0.1	0.00182	[-0.0132, 0.0168]	0.9996
adaptive	seq2seq	1	9.440×10^{-4}	[-0.0141, 0.016]	1.0000
adaptive	seq2seq	10	0.005731	[-0.00928, 0.0207]	0.8231
adaptive	seq2seq	100	0.004947	[-0.0101, 0.02]	0.8999
adaptive	seq2seq	1000	0.01348	[-0.00154, 0.0285]	0.0930
adaptive	seq2seq	1e+04	0.001255	[-0.0138, 0.0163]	1.0000
tropical	pointer	0.01	0.006744	[-0.0427, 0.0562]	0.9882
tropical	pointer	0.1	0.0114	[-0.0381, 0.0609]	0.9270
tropical	pointer	10	0.005948	[-0.0435, 0.0554]	0.9926
tropical	pointer	1000	-0.02035	[-0.0742, 0.0335]	0.7097
tropical	seq2seq	0.01	0.009397	[-0.0153, 0.0341]	0.8249
tropical	seq2seq	0.1	0.0153	[-0.00938, 0.04]	0.3827
tropical	seq2seq	1	0.008052	[-0.0166, 0.0327]	0.9041
tropical	seq2seq	10	0.008799	[-0.0159, 0.0335]	0.8629
tropical	seq2seq	100	0.0216	[-0.00309, 0.0463]	0.1062
tropical	seq2seq	1000	0.01406	[-0.0106, 0.0387]	0.4690
tropical	seq2seq	1e+04	-0.01993	[-0.0446, 0.00475]	0.1544
vanilla	pointer	0.01	0.005723	[-0.00712, 0.0186]	0.7075
vanilla	pointer	0.1	0.004625	[-0.00822, 0.0175]	0.8575
vanilla	pointer	1	0.005547	[-0.0073, 0.0184]	0.7337
vanilla	pointer	10	0.004868	[-0.00798, 0.0177]	0.8276
vanilla	pointer	100	0.004085	[-0.00876, 0.0169]	0.9138
vanilla	pointer	1000	0.005777	[-0.00707, 0.0186]	0.6995
vanilla	pointer	1e+04	-0.002669	[-0.0155, 0.0102]	0.9898
vanilla	seq2seq	0.01	8.942×10^{-4}	[-0.00678, 0.00857]	0.9997

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	seq2seq	0.1	0.002226	[-0.00545, 0.0099]	0.9430
vanilla	seq2seq	1	0.001625	[-0.00605, 0.0093]	0.9886
vanilla	seq2seq	10	0.003893	[-0.00378, 0.0116]	0.5871
vanilla	seq2seq	100	0.004989	[-0.00269, 0.0127]	0.3359
vanilla	seq2seq	1000	0.002072	[-0.0056, 0.00975]	0.9594
vanilla	seq2seq	1e+04	0.006049	[-0.00163, 0.0137]	0.1711

Table 3: `ood_len_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.002529	[0.0011, 0.00408]	0.4610	0.227	0.0928
adaptive	seq2seq	0.001574	$[7.00 \times 10^{-4}$, 0.00275]	0.0461	0.318	0.0190
tropical	pointer	-0.005129	[-0.0111, 9.70×10^{-4}]	0.1365	-0.214	0.3931
tropical	seq2seq	-0.002747	[-0.00491, -4.73×10^{-4}]	0.0569	-0.25	0.0584
vanilla	pointer	-8.691×10^{-4}	[-0.00246, 5.01×10^{-4}]	0.0906	-0.253	0.0588
vanilla	seq2seq	6.615×10^{-4}	[-5.71×10^{-6}, 0.00139]	0.0633	0.322	0.0120

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.01354	[-0.055, 0.0279]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.01557	[-0.033, 0.00183]	0.4563	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.007895	[-0.047, 0.0312]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.01327	[-0.0503, 0.0237]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.0406	[-0.147, 0.0657]	0.9681	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.02826	[-0.0553, -0.00119]	0.3564	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.01975	[-0.0223, -0.0172]	0.0003	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.01518	[-0.0292, -0.00117]	0.3564	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.02183	[-0.0275, -0.0162]	0.0051	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.02335	[-0.0294, -0.0173]	0.0051	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.01165	[-0.032, 0.00866]	0.9326	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.01627	[-0.0352, 0.0027]	0.4563	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.05424	[-0.0803, -0.0282]	0.0445	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.02	[-0.0778, 0.0378]	1.0000	

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.03092	[-0.0696, 0.00776]	0.7252	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.03637	[-0.107, 0.0343]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.02634	[-0.0765, 0.0239]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.02937	[-0.0709, 0.0122]	0.8473	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.02124	[-0.0696, 0.0272]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.02903	[-0.0986, 0.0405]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.005795	[-0.0471, 0.0355]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.02229	[-0.0365, -0.00803]	0.1471	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.02751	[-0.0316, -0.0235]	0.0007	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.02269	[-0.0263, -0.0191]	0.0009	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.02297	[-0.0283, -0.0176]	0.0039	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.02045	[-0.0389, -0.00198]	0.3341	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.02233	[-0.035, -0.00968]	0.1044	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.01088	[-0.0198, -0.00197]	0.2751	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.02708	[-0.0464, -0.00773]	0.1954	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.007042	[-0.0552, 0.0411]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.00791	[-0.0497, 0.0339]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.0206	[-0.0912, 0.05]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.008775	[-0.0632, 0.0457]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.01727	[-0.0714, 0.0368]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.009143	[-0.058, 0.0397]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.02377	[-0.0926, 0.045]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.003241	[-0.0333, 0.0398]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.005388	[-0.0509, 0.0617]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.003037	[-0.0156, 0.0217]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	2.653×10^{-5}	[-0.0208, 0.0209]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.00275	[-0.0326, 0.0381]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.02215	[-0.129, 0.0843]	1.0000	

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.00933	[-0.0318, 0.0131]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.004501	[-0.0142, 0.00521]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.00693	[-0.0182, 0.00433]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.005407	[-0.0118, 9.74×10^{-4}]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.008354	[-0.0206, 0.00389]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.01023	[-0.0148, -0.00563]	0.0736
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.005624	[-0.0164, 0.00519]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.01805	[-0.0323, -0.00379]	0.4920

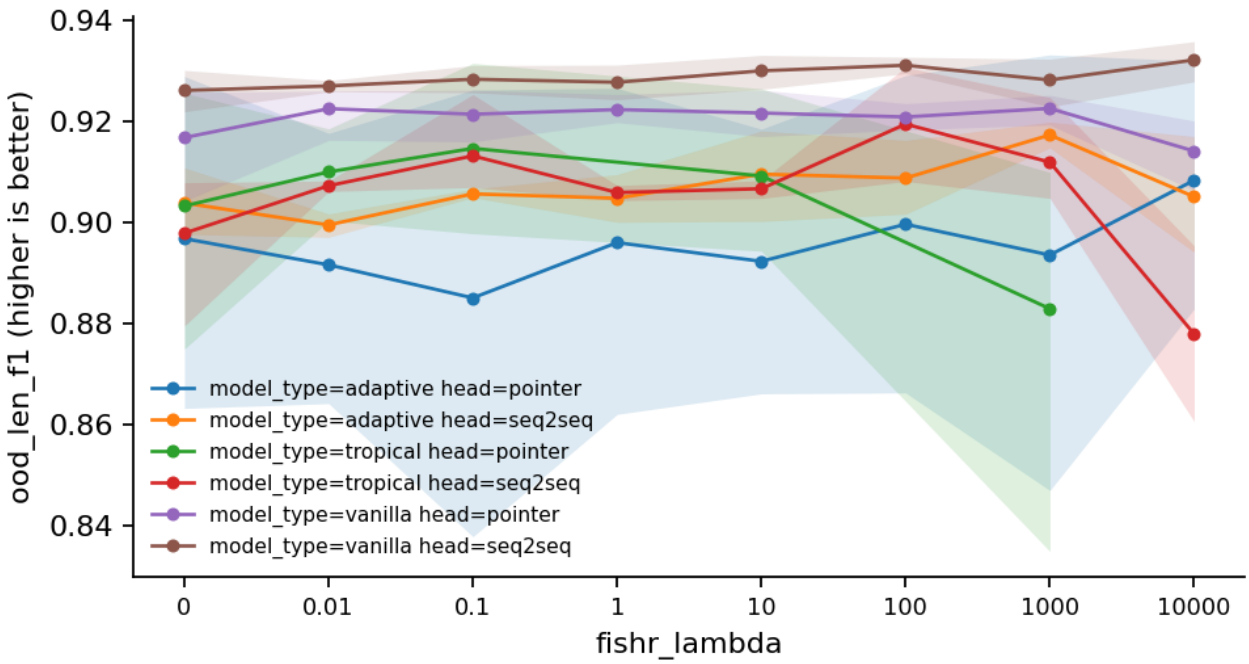


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

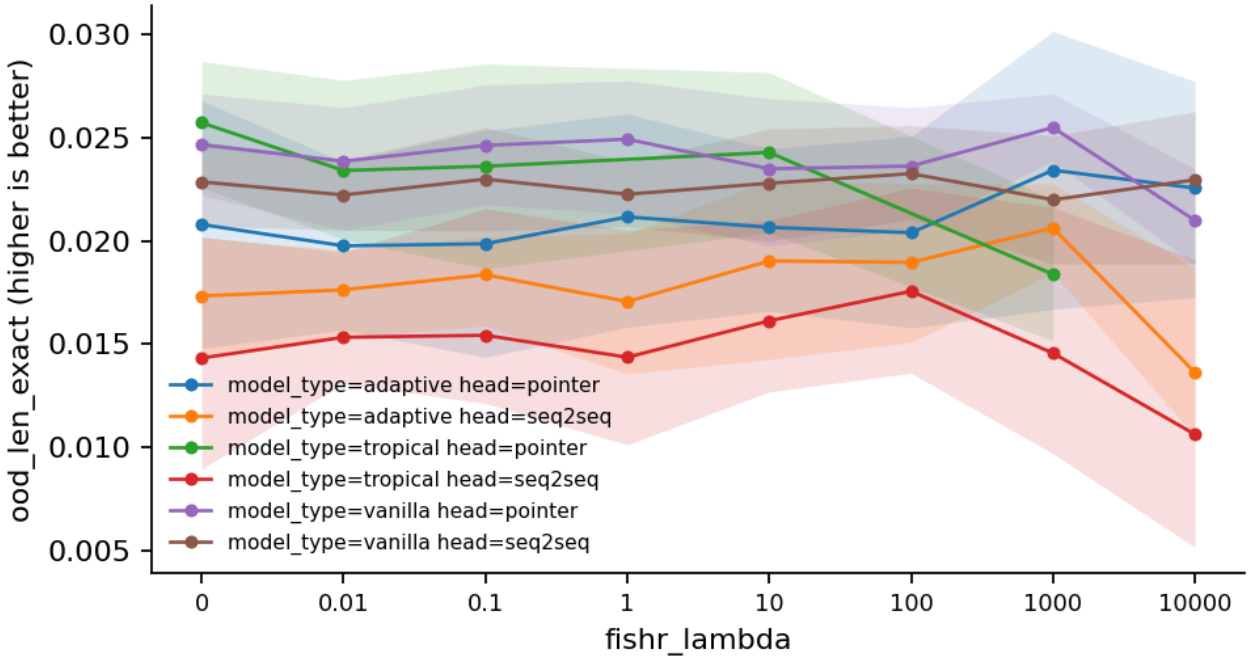


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.0208	[0.0148, 0.0268]	0.00784	5	0
adaptive	pointer	0.01	0.01977	[0.0157, 0.0238]	0.00542	5	0
adaptive	pointer	0.1	0.01987	[0.0144, 0.0254]	0.00711	5	0
adaptive	pointer	1	0.02117	[0.0158, 0.0262]	0.00679	5	0
adaptive	pointer	10	0.02067	[0.0166, 0.0245]	0.0052	5	0
adaptive	pointer	100	0.0204	[0.0158, 0.0251]	0.00602	5	0
adaptive	pointer	1000	0.02343	[0.0167, 0.0302]	0.00901	5	0
adaptive	pointer	1e+04	0.02257	[0.0173, 0.0277]	0.00693	5	0
adaptive	seq2seq	0	0.01733	[0.0145, 0.0202]	0.00385	5	0
adaptive	seq2seq	0.01	0.01763	[0.0153, 0.0196]	0.00279	5	0
adaptive	seq2seq	0.1	0.01837	[0.0159, 0.0202]	0.00289	5	0
adaptive	seq2seq	1	0.01707	[0.0136, 0.0204]	0.00459	5	0
adaptive	seq2seq	10	0.01903	[0.0143, 0.0229]	0.00576	5	0
adaptive	seq2seq	100	0.01897	[0.0151, 0.0228]	0.00497	5	0
adaptive	seq2seq	1000	0.02063	[0.0185, 0.0228]	0.00286	5	0
adaptive	seq2seq	1e+04	0.01363	[0.0105, 0.0187]	0.00568	5	0
tropical	pointer	0	0.02573	[0.0226, 0.0287]	0.00414	5	0
tropical	pointer	0.01	0.02342	[0.0201, 0.0278]	0.00484	4	0
tropical	pointer	0.1	0.02363	[0.0187, 0.0286]	0.0059	4	0
tropical	pointer	10	0.02429	[0.0204, 0.0282]	0.00459	4	0
tropical	pointer	1000	0.01839	[0.0152, 0.022]	0.00343	3	0
tropical	seq2seq	0	0.01433	[0.00897, 0.0202]	0.00669	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.01533	[0.0131, 0.0195]	0.00458	5	0
tropical	seq2seq	0.1	0.01543	[0.0121, 0.0216]	0.00685	5	0
tropical	seq2seq	1	0.01437	[0.0101, 0.0205]	0.00681	5	0
tropical	seq2seq	10	0.01613	[0.0127, 0.021]	0.00535	5	0
tropical	seq2seq	100	0.01757	[0.0136, 0.0226]	0.00579	5	0
tropical	seq2seq	1000	0.01457	[0.0097, 0.0216]	0.00798	5	0
tropical	seq2seq	1e+04	0.01063	[0.00517, 0.0191]	0.0094	5	0
vanilla	pointer	0	0.02467	[0.0222, 0.0271]	0.00313	5	0
vanilla	pointer	0.01	0.02387	[0.0205, 0.0265]	0.00392	5	0
vanilla	pointer	0.1	0.02463	[0.0218, 0.0276]	0.00354	5	0
vanilla	pointer	1	0.02493	[0.0213, 0.0278]	0.00406	5	0
vanilla	pointer	10	0.0235	[0.0198, 0.0269]	0.00461	5	0
vanilla	pointer	100	0.02363	[0.0206, 0.0265]	0.0039	5	0
vanilla	pointer	1000	0.0255	[0.0239, 0.0271]	0.0021	5	0
vanilla	pointer	1e+04	0.021	[0.0189, 0.0235]	0.003	5	0
vanilla	seq2seq	0	0.02287	[0.0207, 0.0248]	0.00272	5	0
vanilla	seq2seq	0.01	0.02223	[0.0205, 0.0239]	0.00221	5	0
vanilla	seq2seq	0.1	0.023	[0.0205, 0.0255]	0.00328	5	0
vanilla	seq2seq	1	0.02227	[0.0207, 0.0238]	0.00207	5	0
vanilla	seq2seq	10	0.0228	[0.02, 0.0254]	0.0037	5	0
vanilla	seq2seq	100	0.02327	[0.0211, 0.0256]	0.00293	5	0
vanilla	seq2seq	1000	0.022	[0.0189, 0.0251]	0.00426	5	0
vanilla	seq2seq	1e+04	0.02297	[0.0189, 0.0263]	0.00456	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.001033	[-0.0131, 0.011]	1.0000
adaptive	pointer	0.1	-9.333×10^{-4}	[-0.013, 0.0111]	1.0000
adaptive	pointer	1	3.667×10^{-4}	[-0.0117, 0.0124]	1.0000
adaptive	pointer	10	-1.333×10^{-4}	[-0.0122, 0.0119]	1.0000
adaptive	pointer	100	-4.000×10^{-4}	[-0.0124, 0.0116]	1.0000
adaptive	pointer	1000	0.002633	[-0.00941, 0.0147]	0.9864
adaptive	pointer	1e+04	0.001767	[-0.0103, 0.0138]	0.9987
adaptive	seq2seq	0.01	3.000×10^{-4}	[-0.00727, 0.00787]	1.0000
adaptive	seq2seq	0.1	0.001033	[-0.00654, 0.00861]	0.9992
adaptive	seq2seq	1	-2.667×10^{-4}	[-0.00784, 0.00731]	1.0000
adaptive	seq2seq	10	0.0017	[-0.00587, 0.00927]	0.9843
adaptive	seq2seq	100	0.001633	[-0.00594, 0.00921]	0.9874
adaptive	seq2seq	1000	0.0033	[-0.00427, 0.0109]	0.7263
adaptive	seq2seq	1e+04	-0.0037	[-0.0113, 0.00387]	0.6236
tropical	pointer	0.01	-0.002317	[-0.011, 0.00633]	0.8840
tropical	pointer	0.1	-0.002108	[-0.0108, 0.00654]	0.9130
tropical	pointer	10	-0.001442	[-0.0101, 0.00721]	0.9757
tropical	pointer	1000	-0.007344	[-0.0168, 0.00207]	0.1521
tropical	seq2seq	0.01	0.001	[-0.0109, 0.0129]	1.0000
tropical	seq2seq	0.1	0.0011	[-0.0108, 0.013]	0.9999
tropical	seq2seq	1	3.333×10^{-5}	[-0.0119, 0.012]	1.0000
tropical	seq2seq	10	0.0018	[-0.0101, 0.0137]	0.9985
tropical	seq2seq	100	0.003233	[-0.0087, 0.0152]	0.9586
tropical	seq2seq	1000	2.333×10^{-4}	[-0.0117, 0.0122]	1.0000
tropical	seq2seq	1e+04	-0.0037	[-0.0156, 0.00823]	0.9228
vanilla	pointer	0.01	-8.000×10^{-4}	[-0.0071, 0.0055]	0.9995

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	pointer	0.1	-3.333×10^{-5}	[-0.00633, 0.00627]	1.0000
vanilla	pointer	1	2.667×10^{-4}	[-0.00603, 0.00657]	1.0000
vanilla	pointer	10	-0.001167	[-0.00747, 0.00513]	0.9947
vanilla	pointer	100	-0.001033	[-0.00733, 0.00527]	0.9974
vanilla	pointer	1000	8.333×10^{-4}	[-0.00547, 0.00713]	0.9993
vanilla	pointer	1e+04	-0.003667	[-0.00997, 0.00263]	0.4468
vanilla	seq2seq	0.01	-6.333×10^{-4}	[-0.00645, 0.00518]	0.9998
vanilla	seq2seq	0.1	1.333×10^{-4}	[-0.00568, 0.00595]	1.0000
vanilla	seq2seq	1	-6.000×10^{-4}	[-0.00641, 0.00521]	0.9999
vanilla	seq2seq	10	-6.667×10^{-5}	[-0.00588, 0.00575]	1.0000
vanilla	seq2seq	100	4.000×10^{-4}	[-0.00541, 0.00621]	1.0000
vanilla	seq2seq	1000	-8.667×10^{-4}	[-0.00668, 0.00495]	0.9986
vanilla	seq2seq	1e+04	1.000×10^{-4}	[-0.00571, 0.00591]	1.0000

Table 7: `ood_len_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	5.274×10^{-4}	$[1.92 \times 10^{-4}, 0.00106]$	0.3260	0.105	0.4309
adaptive	seq2seq	-1.988×10^{-4}	$[-6.99 \times 10^{-4}, 3.52 \times 10^{-4}]$	0.6083	-0.00952	0.9558
tropical	pointer	-8.220×10^{-4}	$[-0.00155, 5.74 \times 10^{-4}]$	0.2424	-0.274	0.2657
tropical	seq2seq	-4.512×10^{-4}	$[-8.65 \times 10^{-4}, 1.18 \times 10^{-5}]$	0.4221	-0.156	0.2412
vanilla	pointer	-2.917×10^{-4}	$[-6.83 \times 10^{-4}, 1.40 \times 10^{-4}]$	0.3454	-0.143	0.2879
vanilla	seq2seq	4.286×10^{-5}	$[-4.56 \times 10^{-4}, 6.42 \times 10^{-4}]$	0.8738	0.0533	0.6961

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.001067	[-0.00417, 0.0063]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.002042	[-0.0106, 0.00649]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.002292	[-0.0125, 0.00791]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-8.674×10^{-19}	[-0.00912, 0.00912]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.007111	[-0.021, 0.0068]	0.9306
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.008533	[-0.0188, 0.00169]	0.7331
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.0069	[-0.0111, -0.00275]	0.1292
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.007567	[-0.0146, -5.33×10^{-4}]	0.4452
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.0079	[-0.0155, -2.93×10^{-4}]	0.4486
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.006667	[-0.0152, 0.00188]	0.7706
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.0057	[-0.0146, 0.0032]	0.9306

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.007433	[-0.0184, 0.00353]	0.9306
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.01233	[-0.0219, -0.00272]	0.2822
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.003867	[-0.0104, 0.00264]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.0041	[-0.0111, 0.00287]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.004767	[-0.0148, 0.00524]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.003767	[-0.0123, 0.00479]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.002833	[-0.00991, 0.00424]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.003233	[-0.00924, 0.00278]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.002067	[-0.0125, 0.00839]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.001567	[-0.00627, 0.0094]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.005533	[-0.0133, 0.00226]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.0046	[-0.00708, -0.00212]	0.1082
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.004633	[-0.00998, 7.10×10^{-4}]	0.9588
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.0052	[-0.00955, -8.53×10^{-4}]	0.4401
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.003767	[-0.0135, 0.00594]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.0043	[-0.0123, 0.00369]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.001367	[-0.00871, 0.00597]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.009333	[-0.0174, -0.00124]	0.4603
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.003467	[-0.00742, 0.0143]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.002133	[-0.00587, 0.0101]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.0015	[-0.00882, 0.0118]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.0041	[-0.00647, 0.0147]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.001633	[-0.00998, 0.0132]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.001433	[-0.00778, 0.0106]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.0028	[-0.00819, 0.0138]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.008933	[-0.00219, 0.0201]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.0114	[7.79×10^{-4}, 0.022]	0.8556
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.007458	[-0.00617, 0.0211]	1.0000

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.007542	[-0.00458, 0.0197]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.007875	$[-9.69 \times 10^{-5}, 0.0158]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.008444	[-0.0033, 0.0202]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.0018	[-0.00398, 0.00758]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.001633	[-0.00401, 0.00728]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.001633	[-0.00428, 0.00755]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.002667	[-0.00238, 0.00771]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	7.000×10^{-4}	[-0.0076, 0.009]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	3.667×10^{-4}	[-0.00471, 0.00545]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.0035	[-0.00324, 0.0102]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.001967	[-0.00787, 0.00394]	1.0000

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.4286	[0.353, 0.524]	0.113	5	0
adaptive	pointer	0.01	0.428	[0.376, 0.48]	0.0685	5	0
adaptive	pointer	0.1	0.4563	[0.41, 0.502]	0.062	5	0
adaptive	pointer	1	0.4473	[0.389, 0.514]	0.0799	5	0
adaptive	pointer	10	0.4328	[0.39, 0.482]	0.0603	5	0
adaptive	pointer	100	0.4348	[0.381, 0.49]	0.0733	5	0
adaptive	pointer	1000	0.4406	[0.387, 0.497]	0.0714	5	0
adaptive	pointer	1e+04	0.4601	[0.395, 0.532]	0.089	5	0
adaptive	seq2seq	0	0.4372	[0.342, 0.534]	0.119	5	0
adaptive	seq2seq	0.01	0.4081	[0.345, 0.46]	0.0758	5	0
adaptive	seq2seq	0.1	0.3908	[0.22, 0.546]	0.199	5	0
adaptive	seq2seq	1	0.4354	[0.318, 0.559]	0.15	5	0
adaptive	seq2seq	10	0.4514	[0.336, 0.572]	0.153	5	0
adaptive	seq2seq	100	0.4562	[0.38, 0.547]	0.111	5	0
adaptive	seq2seq	1000	0.3765	[0.267, 0.461]	0.13	5	0
adaptive	seq2seq	1e+04	0.2804	[0.0807, 0.483]	0.271	5	0
tropical	pointer	0	0.5497	[0.497, 0.613]	0.0743	5	0
tropical	pointer	0.01	0.5453	[0.5, 0.591]	0.0556	4	0
tropical	pointer	0.1	0.5759	[0.523, 0.641]	0.0672	4	0
tropical	pointer	10	0.5096	[0.462, 0.567]	0.0651	4	0
tropical	pointer	1000	0.5272	[0.476, 0.584]	0.0545	3	0
tropical	seq2seq	0	0.2191	[0.0457, 0.392]	0.229	5	0
tropical	seq2seq	0.01	0.1948	[0.0156, 0.396]	0.243	5	0
tropical	seq2seq	0.1	0.2094	[0.0468, 0.378]	0.209	5	0
tropical	seq2seq	1	0.1995	$[9.66 \times 10^{-4}, 0.401]$	0.271	5	0
tropical	seq2seq	10	0.2211	$[6.56 \times 10^{-5}, 0.459]$	0.304	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	100	0.07346	[0, 0.218]	0.162	5	0
tropical	seq2seq	1000	0.1745	[5.64×10^{-4}, 0.373]	0.242	5	0
tropical	seq2seq	1e+04	0.2666	[0.0658, 0.467]	0.254	5	0
vanilla	pointer	0	0.5392	[0.494, 0.585]	0.0596	5	0
vanilla	pointer	0.01	0.5406	[0.518, 0.566]	0.0307	5	0
vanilla	pointer	0.1	0.5207	[0.468, 0.575]	0.0698	5	0
vanilla	pointer	1	0.5596	[0.526, 0.593]	0.0417	5	0
vanilla	pointer	10	0.5216	[0.478, 0.565]	0.0563	5	0
vanilla	pointer	100	0.5572	[0.524, 0.59]	0.0422	5	0
vanilla	pointer	1000	0.5157	[0.451, 0.565]	0.0734	5	0
vanilla	pointer	1e+04	0.5707	[0.552, 0.589]	0.0243	5	0
vanilla	seq2seq	0	0.3531	[0.219, 0.487]	0.168	5	0
vanilla	seq2seq	0.01	0.4509	[0.3, 0.59]	0.195	5	0
vanilla	seq2seq	0.1	0.4106	[0.281, 0.539]	0.175	5	0
vanilla	seq2seq	1	0.4902	[0.341, 0.615]	0.176	5	0
vanilla	seq2seq	10	0.5389	[0.403, 0.644]	0.15	5	0
vanilla	seq2seq	100	0.4338	[0.221, 0.615]	0.251	5	0
vanilla	seq2seq	1000	0.426	[0.199, 0.589]	0.254	5	0
vanilla	seq2seq	1e+04	0.4365	[0.217, 0.558]	0.245	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-6.136×10^{-4}	[-0.138, 0.137]	1.0000
adaptive	pointer	0.1	0.02769	[-0.11, 0.165]	0.9915
adaptive	pointer	1	0.01868	[-0.119, 0.156]	0.9992
adaptive	pointer	10	0.004179	[-0.133, 0.142]	1.0000
adaptive	pointer	100	0.006114	[-0.132, 0.144]	1.0000
adaptive	pointer	1000	0.01198	[-0.126, 0.15]	1.0000
adaptive	pointer	1e+04	0.03143	[-0.106, 0.169]	0.9828
adaptive	seq2seq	0.01	-0.02906	[-0.311, 0.253]	0.9999
adaptive	seq2seq	0.1	-0.04643	[-0.328, 0.236]	0.9974
adaptive	seq2seq	1	-0.001803	[-0.284, 0.28]	1.0000
adaptive	seq2seq	10	0.01421	[-0.268, 0.296]	1.0000
adaptive	seq2seq	100	0.019	[-0.263, 0.301]	1.0000
adaptive	seq2seq	1000	-0.06064	[-0.343, 0.221]	0.9876
adaptive	seq2seq	1e+04	-0.1568	[-0.439, 0.125]	0.4946
tropical	pointer	0.01	-0.004392	[-0.124, 0.116]	0.9999
tropical	pointer	0.1	0.02617	[-0.0939, 0.146]	0.9391
tropical	pointer	10	-0.04016	[-0.16, 0.0799]	0.7847
tropical	pointer	1000	-0.02248	[-0.153, 0.108]	0.9728
tropical	seq2seq	0.01	-0.02435	[-0.448, 0.399]	1.0000
tropical	seq2seq	0.1	-0.009756	[-0.433, 0.414]	1.0000
tropical	seq2seq	1	-0.01967	[-0.443, 0.404]	1.0000
tropical	seq2seq	10	0.001959	[-0.422, 0.425]	1.0000
tropical	seq2seq	100	-0.1457	[-0.569, 0.278]	0.8807
tropical	seq2seq	1000	-0.04469	[-0.468, 0.379]	0.9998
tropical	seq2seq	1e+04	0.04748	[-0.376, 0.471]	0.9998
vanilla	pointer	0.01	0.001428	[-0.0903, 0.0931]	1.0000
vanilla	pointer	0.1	-0.01851	[-0.11, 0.0732]	0.9913
vanilla	pointer	1	0.02042	[-0.0713, 0.112]	0.9849
vanilla	pointer	10	-0.01764	[-0.109, 0.074]	0.9934
vanilla	pointer	100	0.01802	[-0.0737, 0.11]	0.9925

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	pointer	1000	-0.02348	[-0.115, 0.0682]	0.9687
vanilla	pointer	1e+04	0.03152	[-0.0602, 0.123]	0.8810
vanilla	seq2seq	0.01	0.09781	[-0.261, 0.457]	0.9575
vanilla	seq2seq	0.1	0.05755	[-0.301, 0.417]	0.9978
vanilla	seq2seq	1	0.1371	[-0.222, 0.496]	0.8228
vanilla	seq2seq	10	0.1858	[-0.173, 0.545]	0.5671
vanilla	seq2seq	100	0.08075	[-0.278, 0.44]	0.9841
vanilla	seq2seq	1000	0.07292	[-0.286, 0.432]	0.9910
vanilla	seq2seq	1e+04	0.08341	[-0.276, 0.442]	0.9811

Table 11: `ood_val_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.001862	[-0.00256, 0.00545]	0.7474	0.0781	0.5597
adaptive	seq2seq	-0.01395	[-0.0542, 0.0223]	0.3140	-0.139	0.2919
tropical	pointer	-0.008307	[-0.0229, 0.00989]	0.3442	-0.19	0.4451
tropical	seq2seq	7.005×10^{-4}	[-0.00277, 0.00608]	0.9717	-0.0762	0.5693
vanilla	pointer	0.002784	[-0.00238, 0.00795]	0.5267	0.131	0.3381
vanilla	seq2seq	-0.002458	[-0.0251, 0.0185]	0.8843	-0.0114	0.9414

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0105	[-0.146, 0.167]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.004348	[-0.0973, 0.106]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.0754	[-0.124, 0.275]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.02637	[-0.21, 0.157]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.02726	[-0.21, 0.156]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.1339	[-0.343, 0.0748]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.2561	[-0.466, -0.0457]	0.3334	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.2012	[-0.288, -0.114]	0.0396	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.2907	[-0.57, -0.0117]	0.4445	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.3178	[-0.72, 0.0848]	0.7482	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.3604	[-0.67, -0.0505]	0.3522	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.2516	[-0.555, 0.0516]	0.7432	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.1699	[-0.468, 0.129]	1.0000	

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.1106	[-0.247, 0.0262]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.1126	[-0.172, -0.0537]	0.0967
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.06438	[-0.208, 0.0797]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.1123	[-0.202, -0.0224]	0.3848
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.08876	[-0.198, 0.0202]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.1225	[-0.233, -0.0115]	0.4877
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.07512	[-0.223, 0.0729]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.1107	[-0.207, -0.014]	0.4701
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.08411	[-0.228, 0.397]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.04276	[-0.278, 0.193]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01987	[-0.369, 0.329]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.05482	[-0.186, 0.0762]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.08748	[-0.303, 0.128]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.02236	[-0.318, 0.363]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.04946	[-0.454, 0.355]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.1561	[-0.66, 0.348]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.008552	[-0.247, 0.23]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.01989	[-0.0883, 0.128]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.06556	[-0.184, 0.315]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.01193	[-0.129, 0.153]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.01858	[-0.256, 0.219]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.02144	[-0.15, 0.107]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.06407	[-0.162, 0.29]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.1796	[-0.175, 0.534]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.3306	[0.0115, 0.65]	0.8580
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.4372	[0.181, 0.694]	0.2579
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.4329	[0.0713, 0.794]	0.6359
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.2332	[-0.365, 0.832]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.2374	[-0.515, 0.99]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.1861	[-0.0234, 0.396]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.08975	[-0.131, 0.31]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.1101	[-0.0503, 0.27]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.06943	[-0.106, 0.245]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.0173	[-0.205, 0.17]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1234	[-0.173, 0.42]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.08973	[-0.225, 0.405]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.1342	[-0.146, 0.415]	1.0000

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.01054	[0.0036, 0.0234]	0.0143	5	0
adaptive	pointer	0.01	0.00456	[0.00286, 0.00646]	0.00235	5	0
adaptive	pointer	0.1	0.00828	[0.00328, 0.0138]	0.00668	5	0
adaptive	pointer	1	0.00432	[0.00244, 0.0066]	0.00267	5	0
adaptive	pointer	10	0.00664	[0.00566, 0.00786]	0.00146	5	0
adaptive	pointer	100	0.00508	[0.00296, 0.00796]	0.00316	5	0
adaptive	pointer	1000	0.00792	[0.00562, 0.0102]	0.00298	5	0
adaptive	pointer	1e+04	0.0146	[0.00728, 0.0272]	0.0137	5	0
adaptive	seq2seq	0	0.01152	[0.00418, 0.0208]	0.011	5	0
adaptive	seq2seq	0.01	0.00972	[0.00478, 0.0148]	0.00655	5	0
adaptive	seq2seq	0.1	0.00626	[0.00308, 0.00898]	0.00388	5	0
adaptive	seq2seq	1	0.00886	[0.00456, 0.0127]	0.00551	5	0
adaptive	seq2seq	10	0.01562	[0.00348, 0.0284]	0.016	5	0
adaptive	seq2seq	100	0.01244	[0.0051, 0.0219]	0.0111	5	0
adaptive	seq2seq	1000	0.00778	[0.00328, 0.0123]	0.00606	5	0
adaptive	seq2seq	1e+04	0.00122	$[2.00 \times 10^{-5}, 0.0032]$	0.00215	5	0
tropical	pointer	0	0.01174	[0.0046, 0.0189]	0.00939	5	0
tropical	pointer	0.01	0.01553	[0.00525, 0.0347]	0.0191	4	0
tropical	pointer	0.1	0.0174	[0.0028, 0.0406]	0.0233	4	0
tropical	pointer	10	0.0128	[0.00765, 0.018]	0.00638	4	0
tropical	pointer	1000	0.01433	[0.0096, 0.0192]	0.0048	3	0
tropical	seq2seq	0	0.00572	$[2.20 \times 10^{-4}, 0.0151]$	0.0103	5	0
tropical	seq2seq	0.01	0.00608	$[6.00 \times 10^{-5}, 0.0155]$	0.0102	5	0
tropical	seq2seq	0.1	0.00396	$[2.40 \times 10^{-4}, 0.00916]$	0.00576	5	0
tropical	seq2seq	1	0.0066	[0, 0.0138]	0.0091	5	0
tropical	seq2seq	10	0.01146	[0, 0.0245]	0.0159	5	0
tropical	seq2seq	100	0.00104	[0, 0.00312]	0.00233	5	0
tropical	seq2seq	1000	0.0048	$[1.00 \times 10^{-4}, 0.0109]$	0.00708	5	0
tropical	seq2seq	1e+04	0.00808	$[8.00 \times 10^{-4}, 0.0158]$	0.00962	5	0
vanilla	pointer	0	0.012	[0.00224, 0.0259]	0.0154	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
vanilla	pointer	0.01	0.01388	[0.00266, 0.0283]	0.0159	5	0
vanilla	pointer	0.1	0.01464	[0.00536, 0.0239]	0.0121	5	0
vanilla	pointer	1	0.01664	[9.00×10^{-4}, 0.0401]	0.0255	5	0
vanilla	pointer	10	0.01152	[0.0014, 0.0233]	0.0141	5	0
vanilla	pointer	100	0.0126	[0.00232, 0.0259]	0.0152	5	0
vanilla	pointer	1000	0.0231	[0.00962, 0.0366]	0.0174	5	0
vanilla	pointer	1e+04	0.0145	[0.00232, 0.0321]	0.0195	5	0
vanilla	seq2seq	0	0.00974	[0.00148, 0.018]	0.0108	5	0
vanilla	seq2seq	0.01	0.01544	[0.00508, 0.0258]	0.0133	5	0
vanilla	seq2seq	0.1	0.01388	[0.00446, 0.0233]	0.0128	5	0
vanilla	seq2seq	1	0.01746	[0.00696, 0.028]	0.0135	5	0
vanilla	seq2seq	10	0.0149	[0.00354, 0.0263]	0.0143	5	0
vanilla	seq2seq	100	0.01908	[0.0066, 0.0316]	0.0169	5	0
vanilla	seq2seq	1000	0.0097	[0.00238, 0.017]	0.00929	5	0
vanilla	seq2seq	1e+04	0.02048	[0.01, 0.0288]	0.0127	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-6.000×10^{-4}	[-0.0193, 0.0019]	1.0000
adaptive	pointer	0.1	2.000×10^{-4}	[-0.0177, 0.00936]	1.0000
adaptive	pointer	1	0	[-0.0193, 0.00172]	1.0000
adaptive	pointer	10	0.0017	[-0.0174, 0.00402]	1.0000
adaptive	pointer	100	-6.000×10^{-4}	[-0.0182, 0.00262]	1.0000
adaptive	pointer	1000	0.0016	[-0.0153, 0.00526]	1.0000
adaptive	pointer	1e+04	0.0027	[0.00262, 0.00674]	0.4375
adaptive	seq2seq	0.01	-0.0019	[-0.00628, 0.0023]	1.0000
adaptive	seq2seq	0.1	-0.003	[-0.0127, 0.00102]	1.0000
adaptive	seq2seq	1	-1.000×10^{-4}	[-0.01, 0.00354]	1.0000
adaptive	seq2seq	10	-5.000×10^{-4}	[-0.00382, 0.0153]	1.0000
adaptive	seq2seq	100	7.000×10^{-4}	[-0.00854, 0.0127]	1.0000
adaptive	seq2seq	1000	-3.000×10^{-4}	[-0.017, 0.00478]	1.0000
adaptive	seq2seq	1e+04	-0.0047	[-0.0207, -0.00306]	0.4375
tropical	pointer	0.01	-0.0042	[-0.0153, 0.0222]	1.0000
tropical	pointer	0.1	-0.00595	[-0.0123, 0.0285]	1.0000
tropical	pointer	10	-0.00125	[-0.00775, 0.0107]	1.0000
tropical	pointer	1000	-0.0091	[-0.0094, 0.0127]	1.0000
tropical	seq2seq	0.01	0	[-0.0105, 0.0119]	1.0000
tropical	seq2seq	0.1	0	[-0.0112, 0.00598]	1.0000
tropical	seq2seq	1	0	[-0.0036, 0.00668]	1.0000
tropical	seq2seq	10	0	[-6.60×10^{-4}, 0.0174]	1.0000
tropical	seq2seq	100	0	[-0.0144, 8.00×10^{-4}]	1.0000
tropical	seq2seq	1000	0	[-0.00992, 0.0075]	1.0000
tropical	seq2seq	1e+04	0	[-0.00412, 0.0106]	1.0000
vanilla	pointer	0.01	5.000×10^{-4}	[-0.00908, 0.0146]	1.0000
vanilla	pointer	0.1	0	[-0.00918, 0.0146]	1.0000
vanilla	pointer	1	-0.0012	[-0.0109, 0.0268]	1.0000
vanilla	pointer	10	-5.000×10^{-4}	[-0.00952, 0.00988]	1.0000
vanilla	pointer	100	3.000×10^{-4}	[-0.0121, 0.0138]	1.0000
vanilla	pointer	1000	0.0058	[0.0028, 0.0214]	0.4375
vanilla	pointer	1e+04	0.001	[6.00×10^{-5}, 0.00606]	1.0000
vanilla	seq2seq	0.01	0.0011	[-0.00124, 0.0152]	1.0000

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
vanilla	seq2seq	0.1	-6.000×10^{-4}	[-0.00188, 0.0114]	1.0000
vanilla	seq2seq	1	0.0067	$[-9.80 \times 10^{-4}$, 0.0194]	1.0000
vanilla	seq2seq	10	-0.0013	[-0.0124, 0.0227]	1.0000
vanilla	seq2seq	100	0.0091	$[5.00 \times 10^{-4}$, 0.0207]	1.0000
vanilla	seq2seq	1000	-0.0038	[-0.0116, 0.0127]	1.0000
vanilla	seq2seq	1e+04	0.004	$[5.20 \times 10^{-4}$, 0.0237]	1.0000

Table 15: `ood_val_exact` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.4089	[0.0439, 0.721]	0.0147	0.33	0.0138
adaptive	seq2seq	-0.3049	[-0.54, 0.0384]	0.0749	-0.24	0.0640
tropical	pointer	0.3132	[-0.199, 0.855]	0.2556	0.262	0.2877
tropical	seq2seq	-0.02856	[-0.0731, 0.0398]	0.8706	-0.0152	0.9084
vanilla	pointer	0.09411	[-0.123, 0.399]	0.5908	0.0762	0.5677
vanilla	seq2seq	-7.083×10^{-4}	[-0.294, 0.297]	0.9968	0.0019	0.9964

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-2.600×10^{-4}	[-0.0267, 0.0262]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.00175	[-0.0506, 0.0471]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-5.750×10^{-4}	[-0.0371, 0.0359]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.001025	[-0.0151, 0.0131]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.008633	[-0.0553, 0.038]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.00402	[-0.0141, 0.00608]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.00936	[-0.0215, 0.00276]	0.8125
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.00992	[-0.02, 1.28×10^{-4}]	0.8125
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.01086	[-0.0269, 0.00518]	0.8125
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.00344	[-0.0358, 0.0289]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.01804	[-0.0381, 0.002]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.0049	[-0.0182, 0.00835]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.0124	[-0.0288, 0.00399]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.00146	[-0.0313, 0.0284]	1.0000

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.00932	[-0.0278, 0.0092]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.00636	[-0.0173, 0.00459]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.01232	[-0.0412, 0.0166]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.00488	[-0.0211, 0.0113]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.00752	[-0.0233, 0.00829]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.01518	[-0.0361, 0.00578]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	1.000×10^{-4}	[-0.0309, 0.0311]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.00178	[-0.0209, 0.0245]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.00572	[-0.0206, 0.00913]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.00762	[-0.0228, 0.00757]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.0086	[-0.0189, 0.00172]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	7.200×10^{-4}	[-0.00487, 0.00631]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.00664	[-0.037, 0.0237]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.00192	[-0.0178, 0.014]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.01926	[-0.0375, -9.90×10^{-4}]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-9.800×10^{-4}	[-0.0281, 0.0261]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.00516	[-0.0155, 0.00514]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.00202	[-0.00558, 0.00962]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.00454	[-0.0128, 0.00372]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.00898	[-0.0296, 0.0116]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.00736	[-0.0235, 0.0088]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	1.400×10^{-4}	[-0.00937, 0.00965]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.01338	[-0.00427, 0.031]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.00602	[-0.00703, 0.0191]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.01383	[-0.019, 0.0467]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.01582	[-0.0228, 0.0545]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.001525	[-0.038, 0.035]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.0065	[-0.0101, 0.0231]	1.0000

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.00226	[-0.0257, 0.0302]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.00156	[-0.0371, 0.034]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	7.600×10^{-4}	[-0.0268, 0.0283]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-8.200×10^{-4}	[-0.0455, 0.0439]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.00338	[-0.0245, 0.0177]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.00648	[-0.045, 0.032]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.0134	[-0.00896, 0.0358]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.00598	[-0.0456, 0.0336]	1.0000

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.5657	[0.544, 0.588]	0.0287	5	0
adaptive	pointer	0.01	0.5698	[0.552, 0.582]	0.0201	5	0
adaptive	pointer	0.1	0.5722	[0.543, 0.596]	0.0343	5	0
adaptive	pointer	1	0.5711	[0.559, 0.582]	0.0146	5	0
adaptive	pointer	10	0.5509	[0.527, 0.575]	0.0327	5	0
adaptive	pointer	100	0.5657	[0.537, 0.587]	0.033	5	0
adaptive	pointer	1000	0.5595	[0.539, 0.579]	0.0252	5	0
adaptive	pointer	1e+04	0.5743	[0.557, 0.59]	0.0209	5	0
adaptive	seq2seq	0	0.649	[0.628, 0.67]	0.0275	5	0
adaptive	seq2seq	0.01	0.6193	[0.584, 0.643]	0.0407	5	0
adaptive	seq2seq	0.1	0.602	[0.487, 0.676]	0.128	5	0
adaptive	seq2seq	1	0.642	[0.55, 0.708]	0.107	5	0
adaptive	seq2seq	10	0.601	[0.564, 0.638]	0.0469	5	0
adaptive	seq2seq	100	0.6475	[0.613, 0.681]	0.0464	5	0
adaptive	seq2seq	1000	0.6521	[0.614, 0.68]	0.0446	5	0
adaptive	seq2seq	1e+04	0.4993	[0.344, 0.634]	0.195	5	0
tropical	pointer	0	0.5883	[0.566, 0.61]	0.0295	5	0
tropical	pointer	0.01	0.5928	[0.574, 0.615]	0.0241	4	0
tropical	pointer	0.1	0.5719	[0.556, 0.583]	0.0163	4	0
tropical	pointer	10	0.5568	[0.535, 0.579]	0.0262	4	0
tropical	pointer	1000	0.6148	[0.598, 0.636]	0.0192	3	0
tropical	seq2seq	0	0.4824	[0.313, 0.584]	0.189	5	0
tropical	seq2seq	0.01	0.3977	[0.201, 0.556]	0.217	5	0
tropical	seq2seq	0.1	0.4676	[0.27, 0.603]	0.223	5	0
tropical	seq2seq	1	0.3701	[0.149, 0.595]	0.293	5	0
tropical	seq2seq	10	0.3694	[0.142, 0.597]	0.289	5	0
tropical	seq2seq	100	0.3017	[0.107, 0.494]	0.254	5	0
tropical	seq2seq	1000	0.354	[0.19, 0.509]	0.218	5	0
tropical	seq2seq	1e+04	0.4491	[0.235, 0.608]	0.248	5	0
vanilla	pointer	0	0.5198	[0.501, 0.538]	0.0239	5	0
vanilla	pointer	0.01	0.5431	[0.493, 0.586]	0.0603	5	0
vanilla	pointer	0.1	0.5592	[0.539, 0.572]	0.0228	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
vanilla	pointer	1	0.5383	[0.516, 0.575]	0.0387	5	0
vanilla	pointer	10	0.5674	[0.518, 0.605]	0.058	5	0
vanilla	pointer	100	0.5353	[0.493, 0.578]	0.055	5	0
vanilla	pointer	1000	0.564	[0.522, 0.601]	0.0501	5	0
vanilla	pointer	1e+04	0.5428	[0.511, 0.575]	0.0422	5	0
vanilla	seq2seq	0	0.6124	[0.582, 0.644]	0.0403	5	0
vanilla	seq2seq	0.01	0.6103	[0.537, 0.666]	0.0836	5	0
vanilla	seq2seq	0.1	0.6215	[0.588, 0.651]	0.0408	5	0
vanilla	seq2seq	1	0.6391	[0.588, 0.676]	0.0563	5	0
vanilla	seq2seq	10	0.6545	[0.621, 0.685]	0.0435	5	0
vanilla	seq2seq	100	0.5999	[0.531, 0.654]	0.0817	5	0
vanilla	seq2seq	1000	0.5765	[0.486, 0.66]	0.109	5	0
vanilla	seq2seq	1e+04	0.5354	[0.385, 0.622]	0.165	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.004067	[-0.0431, 0.0513]	1.0000
adaptive	pointer	0.1	0.00648	[-0.0407, 0.0537]	0.9991
adaptive	pointer	1	0.00536	[-0.0418, 0.0525]	0.9997
adaptive	pointer	10	-0.01483	[-0.062, 0.0324]	0.9181
adaptive	pointer	100	1.744×10^{-5}	[-0.0472, 0.0472]	1.0000
adaptive	pointer	1000	-0.006204	[-0.0534, 0.041]	0.9994
adaptive	pointer	1e+04	0.008623	[-0.0386, 0.0558]	0.9951
adaptive	seq2seq	0.01	-0.02972	[-0.198, 0.139]	0.9961
adaptive	seq2seq	0.1	-0.04704	[-0.216, 0.122]	0.9526
adaptive	seq2seq	1	-0.007006	[-0.176, 0.162]	1.0000
adaptive	seq2seq	10	-0.04801	[-0.217, 0.121]	0.9478
adaptive	seq2seq	100	-0.001531	[-0.17, 0.167]	1.0000
adaptive	seq2seq	1000	0.003071	[-0.166, 0.172]	1.0000
adaptive	seq2seq	1e+04	-0.1497	[-0.318, 0.0191]	0.0990
tropical	pointer	0.01	0.004452	[-0.0403, 0.0492]	0.9964
tropical	pointer	0.1	-0.01647	[-0.0612, 0.0283]	0.7270
tropical	pointer	10	-0.03152	[-0.0763, 0.0132]	0.2157
tropical	pointer	1000	0.02645	[-0.0223, 0.0752]	0.4214
tropical	seq2seq	0.01	-0.0847	[-0.511, 0.341]	0.9920
tropical	seq2seq	0.1	-0.01472	[-0.441, 0.411]	1.0000
tropical	seq2seq	1	-0.1123	[-0.538, 0.314]	0.9638
tropical	seq2seq	10	-0.113	[-0.539, 0.313]	0.9627
tropical	seq2seq	100	-0.1807	[-0.606, 0.245]	0.7479
tropical	seq2seq	1000	-0.1284	[-0.554, 0.297]	0.9319
tropical	seq2seq	1e+04	-0.03325	[-0.459, 0.393]	1.0000
vanilla	pointer	0.01	0.02329	[-0.057, 0.104]	0.9428
vanilla	pointer	0.1	0.03942	[-0.0409, 0.12]	0.6189
vanilla	pointer	1	0.01856	[-0.0617, 0.0988]	0.9816
vanilla	pointer	10	0.04759	[-0.0327, 0.128]	0.4283
vanilla	pointer	100	0.01556	[-0.0647, 0.0958]	0.9931
vanilla	pointer	1000	0.04425	[-0.036, 0.125]	0.5031
vanilla	pointer	1e+04	0.02299	[-0.0573, 0.103]	0.9463
vanilla	seq2seq	0.01	-0.002057	[-0.155, 0.151]	1.0000
vanilla	seq2seq	0.1	0.009067	[-0.144, 0.162]	1.0000
vanilla	seq2seq	1	0.0267	[-0.126, 0.179]	0.9962
vanilla	seq2seq	10	0.04212	[-0.111, 0.195]	0.9548

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	seq2seq	100	-0.01249	[-0.165, 0.14]	1.0000
vanilla	seq2seq	1000	-0.03588	[-0.189, 0.117]	0.9799
vanilla	seq2seq	1e+04	-0.07696	[-0.23, 0.0757]	0.5932

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-6.086×10^{-4}	[-0.00414, 0.00165]	0.7825	-0.0438	0.7530
adaptive	seq2seq	-0.009077	[-0.0333, 0.011]	0.3161	-0.0476	0.7355
tropical	pointer	0.002916	[-0.00658, 0.00943]	0.4973	0.0476	0.8868
tropical	seq2seq	-0.005046	[-0.0232, 0.0131]	0.8026	-0.0324	0.8206
vanilla	pointer	2.052×10^{-4}	[-0.00625, 0.00666]	0.9584	0.00571	0.9792
vanilla	seq2seq	-0.01264	[-0.0271, -0.00182]	0.1037	-0.189	0.1682

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.06857	[0.0209, 0.116]	0.2101
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.0616	[-0.0606, 0.184]	0.9910
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.01632	[-0.0405, 0.0731]	0.9910
model_type: tropical - vanilla head=pointer fishr_lambda=10	-7.843×10^{-4}	[-0.0902, 0.0886]	0.9910
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.0621	[-0.0225, 0.147]	0.7529
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.13	[-0.385, 0.125]	0.9910
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.2127	[-0.383, -0.042]	0.3100
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.1538	[-0.406, 0.098]	0.9910
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.269	[-0.6, 0.0624]	0.7529
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.2851	[-0.63, 0.0598]	0.7529
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.2982	[-0.589, -0.00777]	0.5100
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.2225	[-0.481, 0.0361]	0.7529
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.08633	[-0.233, 0.06]	0.9910
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.04592	[-0.00583, 0.0977]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.0267	[-0.0617, 0.115]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.01298	[-0.0499, 0.0758]	1.0000

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.03272	[-0.0145, 0.0799]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.0165	[-0.107, 0.0739]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.03037	[-0.0266, 0.0873]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.004534	[-0.0766, 0.0676]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.03156	[-0.0172, 0.0803]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.03659	[-0.0282, 0.101]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.00893	[-0.104, 0.122]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01951	[-0.158, 0.119]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.002889	[-0.0781, 0.0839]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.05354	[-0.102, -0.00512]	0.5968
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.04755	[-0.0951, 0.19]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.07555	[-0.0655, 0.217]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.03612	[-0.402, 0.329]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.0833	[-0.139, -0.0276]	0.2705
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.04951	[-0.108, 0.00943]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.02979	[-0.155, 0.0954]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.07094	[-0.204, 0.0621]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.05013	[-0.105, 0.00495]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.08175	[-0.174, 0.0107]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.09258	[-0.149, -0.0364]	0.2039
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.07499	[-0.185, 0.335]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.106	[-0.109, 0.321]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.2565	[-0.0283, 0.541]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.1439	[-0.207, 0.494]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.1351	[-0.33, 0.6]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.2268	[-0.509, 0.963]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.09263	[-0.143, -0.0426]	0.1428
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.06728	[-0.208, 0.0735]	1.0000

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.06227	[-0.129, 0.00396]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.1008	[-0.187, -0.0144]	0.5715
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.08716	[-0.193, 0.0187]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.06457	[-0.149, 0.0194]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.0125	[-0.143, 0.118]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.007316	[-0.185, 0.199]	1.0000

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.04214	[0.039, 0.0455]	0.00391	5	0
adaptive	pointer	0.01	0.0413	[0.0398, 0.043]	0.00205	5	0
adaptive	pointer	0.1	0.03832	[0.0358, 0.0408]	0.00331	5	0
adaptive	pointer	1	0.04088	[0.0398, 0.042]	0.00143	5	0
adaptive	pointer	10	0.03826	[0.0338, 0.0414]	0.00508	5	0
adaptive	pointer	100	0.0394	[0.0338, 0.0434]	0.00638	5	0
adaptive	pointer	1000	0.03994	[0.0365, 0.0434]	0.00457	5	0
adaptive	pointer	1e+04	0.04018	[0.0359, 0.0432]	0.0048	5	0
adaptive	seq2seq	0	0.03142	[0.0271, 0.0359]	0.00569	5	0
adaptive	seq2seq	0.01	0.02844	[0.0161, 0.0354]	0.0136	5	0
adaptive	seq2seq	0.1	0.02702	[0.0196, 0.0339]	0.00958	5	0
adaptive	seq2seq	1	0.01992	[0.0108, 0.029]	0.0117	5	0
adaptive	seq2seq	10	0.02694	[0.0177, 0.0352]	0.0117	5	0
adaptive	seq2seq	100	0.02882	[0.0202, 0.0366]	0.0104	5	0
adaptive	seq2seq	1000	0.02248	[0.015, 0.0308]	0.0101	5	0
adaptive	seq2seq	1e+04	0.02008	[0.0114, 0.0288]	0.0114	5	0
tropical	pointer	0	0.03784	[0.0334, 0.0422]	0.00533	5	0
tropical	pointer	0.01	0.041	[0.0363, 0.0457]	0.00577	4	0
tropical	pointer	0.1	0.03775	[0.0312, 0.0422]	0.00665	4	0
tropical	pointer	10	0.03648	[0.0307, 0.0422]	0.00692	4	0
tropical	pointer	1000	0.0374	[0.0329, 0.0398]	0.0039	3	0
tropical	seq2seq	0	0.02706	[0.0154, 0.035]	0.013	5	0
tropical	seq2seq	0.01	0.02082	[0.00998, 0.0309]	0.0133	5	0
tropical	seq2seq	0.1	0.0248	[0.0129, 0.0341]	0.0137	5	0
tropical	seq2seq	1	0.0206	[0.00816, 0.0327]	0.0161	5	0
tropical	seq2seq	10	0.01494	[0.00656, 0.0247]	0.0115	5	0
tropical	seq2seq	100	0.01732	[0.00604, 0.0288]	0.0148	5	0
tropical	seq2seq	1000	0.01818	[0.00874, 0.0282]	0.0129	5	0
tropical	seq2seq	1e+04	0.02258	[0.0128, 0.0318]	0.0129	5	0
vanilla	pointer	0	0.03204	[0.0292, 0.0352]	0.00383	5	0
vanilla	pointer	0.01	0.03248	[0.0306, 0.0349]	0.00279	5	0
vanilla	pointer	0.1	0.03418	[0.0301, 0.0382]	0.00541	5	0
vanilla	pointer	1	0.03418	[0.0305, 0.0379]	0.00468	5	0
vanilla	pointer	10	0.03626	[0.0317, 0.0411]	0.0061	5	0
vanilla	pointer	100	0.03228	[0.0277, 0.0368]	0.00599	5	0
vanilla	pointer	1000	0.03742	[0.031, 0.0422]	0.00717	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
vanilla	pointer	1e+04	0.03174	[0.0294, 0.0347]	0.00362	5	0
vanilla	seq2seq	0	0.03438	[0.0314, 0.0374]	0.00418	5	0
vanilla	seq2seq	0.01	0.02788	[0.0203, 0.0355]	0.00982	5	0
vanilla	seq2seq	0.1	0.03264	[0.0303, 0.0355]	0.00331	5	0
vanilla	seq2seq	1	0.02736	[0.0197, 0.0326]	0.00858	5	0
vanilla	seq2seq	10	0.02662	[0.0167, 0.0359]	0.0135	5	0
vanilla	seq2seq	100	0.02854	[0.0203, 0.0342]	0.00925	5	0
vanilla	seq2seq	1000	0.0232	[0.0105, 0.0351]	0.0164	5	0
vanilla	seq2seq	1e+04	0.03024	[0.0201, 0.0365]	0.0114	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-8.400×10^{-4}	[-0.00822, 0.00654]	0.9997
adaptive	pointer	0.1	-0.00382	[-0.0112, 0.00356]	0.5673
adaptive	pointer	1	-0.00126	[-0.00864, 0.00612]	0.9967
adaptive	pointer	10	-0.00388	[-0.0113, 0.0035]	0.5514
adaptive	pointer	100	-0.00274	[-0.0101, 0.00464]	0.8402
adaptive	pointer	1000	-0.0022	[-0.00958, 0.00518]	0.9353
adaptive	pointer	1e+04	-0.00196	[-0.00934, 0.00542]	0.9626
adaptive	seq2seq	0.01	-0.00298	[-0.0218, 0.0158]	0.9979
adaptive	seq2seq	0.1	-0.0044	[-0.0232, 0.0144]	0.9804
adaptive	seq2seq	1	-0.0115	[-0.0303, 0.0073]	0.3962
adaptive	seq2seq	10	-0.00448	[-0.0233, 0.0143]	0.9784
adaptive	seq2seq	100	-0.0026	[-0.0214, 0.0162]	0.9991
adaptive	seq2seq	1000	-0.00894	[-0.0277, 0.00986]	0.6492
adaptive	seq2seq	1e+04	-0.01134	[-0.0301, 0.00746]	0.4104
tropical	pointer	0.01	0.00316	[-0.0077, 0.014]	0.8524
tropical	pointer	0.1	-9.000×10^{-5}	[-0.011, 0.0108]	1.0000
tropical	pointer	10	-0.001365	[-0.0122, 0.0095]	0.9913
tropical	pointer	1000	-4.400×10^{-4}	[-0.0123, 0.0114]	0.9999
tropical	seq2seq	0.01	-0.00624	[-0.03, 0.0175]	0.9645
tropical	seq2seq	0.1	-0.00226	[-0.026, 0.0215]	0.9999
tropical	seq2seq	1	-0.00646	[-0.0302, 0.0173]	0.9579
tropical	seq2seq	10	-0.01212	[-0.0359, 0.0116]	0.5813
tropical	seq2seq	100	-0.00974	[-0.0335, 0.014]	0.7741
tropical	seq2seq	1000	-0.00888	[-0.0326, 0.0149]	0.8361
tropical	seq2seq	1e+04	-0.00448	[-0.0282, 0.0193]	0.9941
vanilla	pointer	0.01	4.400×10^{-4}	[-0.00854, 0.00942]	1.0000
vanilla	pointer	0.1	0.00214	[-0.00684, 0.0111]	0.9784
vanilla	pointer	1	0.00214	[-0.00684, 0.0111]	0.9783
vanilla	pointer	10	0.00422	[-0.00476, 0.0132]	0.6597
vanilla	pointer	100	2.400×10^{-4}	[-0.00874, 0.00922]	1.0000
vanilla	pointer	1000	0.00538	[-0.0036, 0.0144]	0.4168
vanilla	pointer	1e+04	-3.000×10^{-4}	[-0.00928, 0.00868]	1.0000
vanilla	seq2seq	0.01	-0.0065	[-0.0247, 0.0117]	0.8610
vanilla	seq2seq	0.1	-0.00174	[-0.0199, 0.0164]	0.9999
vanilla	seq2seq	1	-0.00702	[-0.0252, 0.0112]	0.8155
vanilla	seq2seq	10	-0.00776	[-0.0259, 0.0104]	0.7427
vanilla	seq2seq	100	-0.00584	[-0.024, 0.0123]	0.9099
vanilla	seq2seq	1000	-0.01118	[-0.0294, 0.00699]	0.3903
vanilla	seq2seq	1e+04	-0.00414	[-0.0223, 0.014]	0.9830

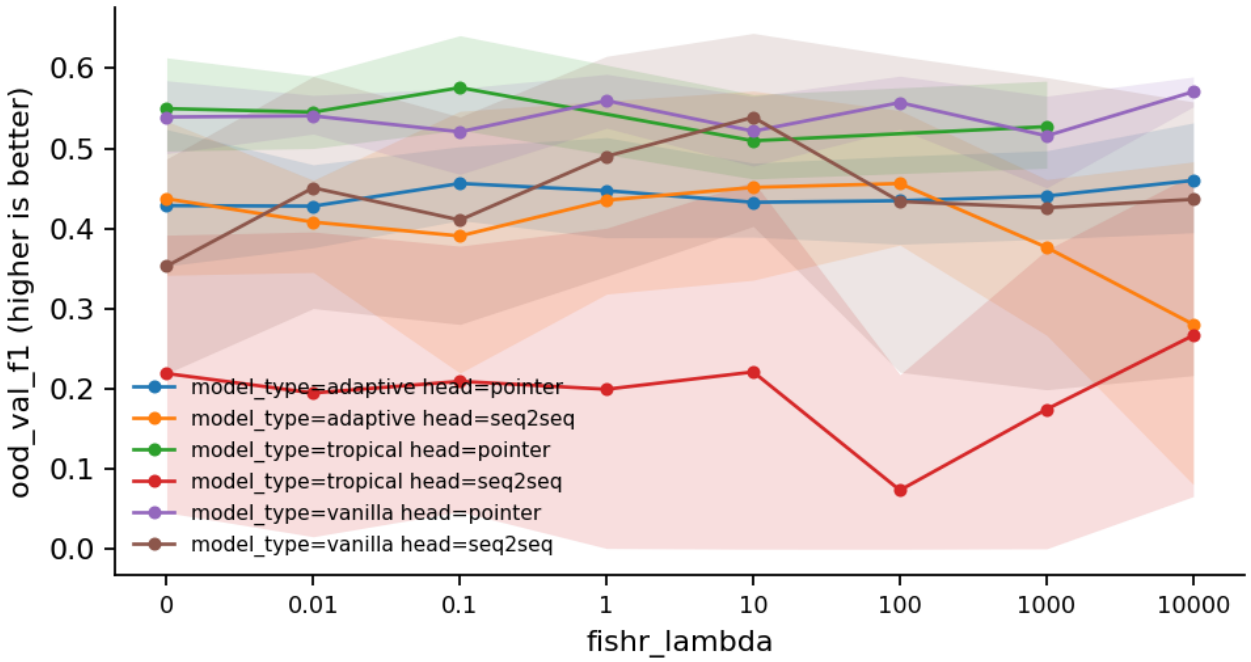


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

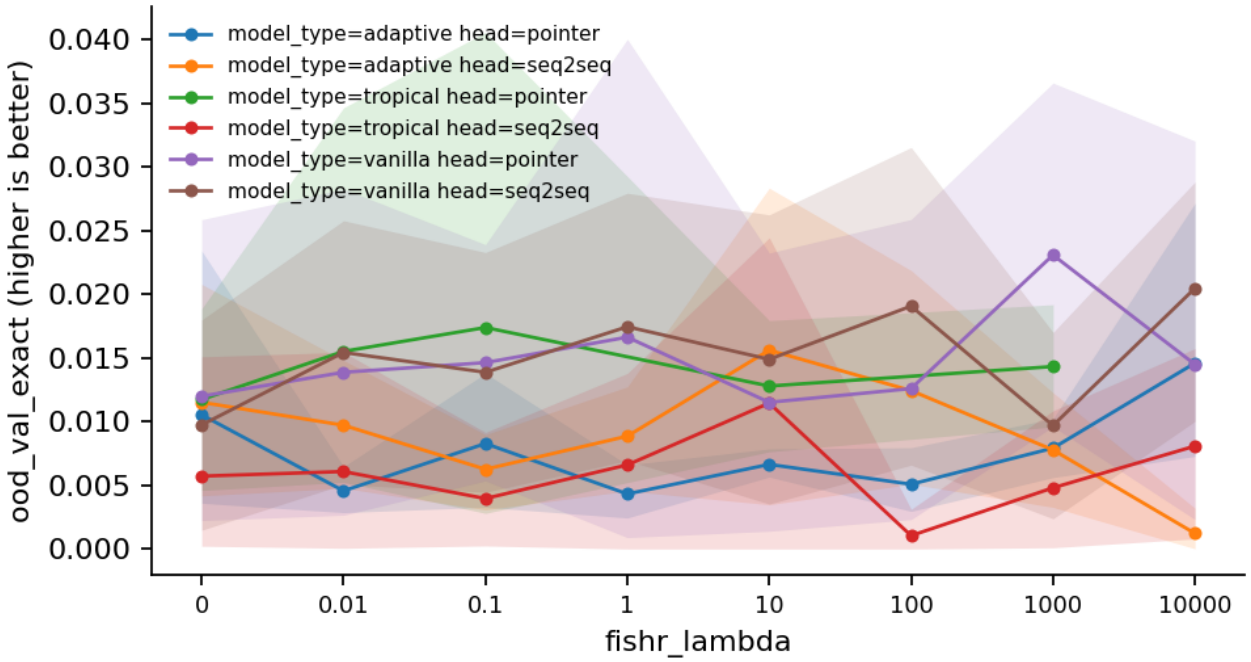


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

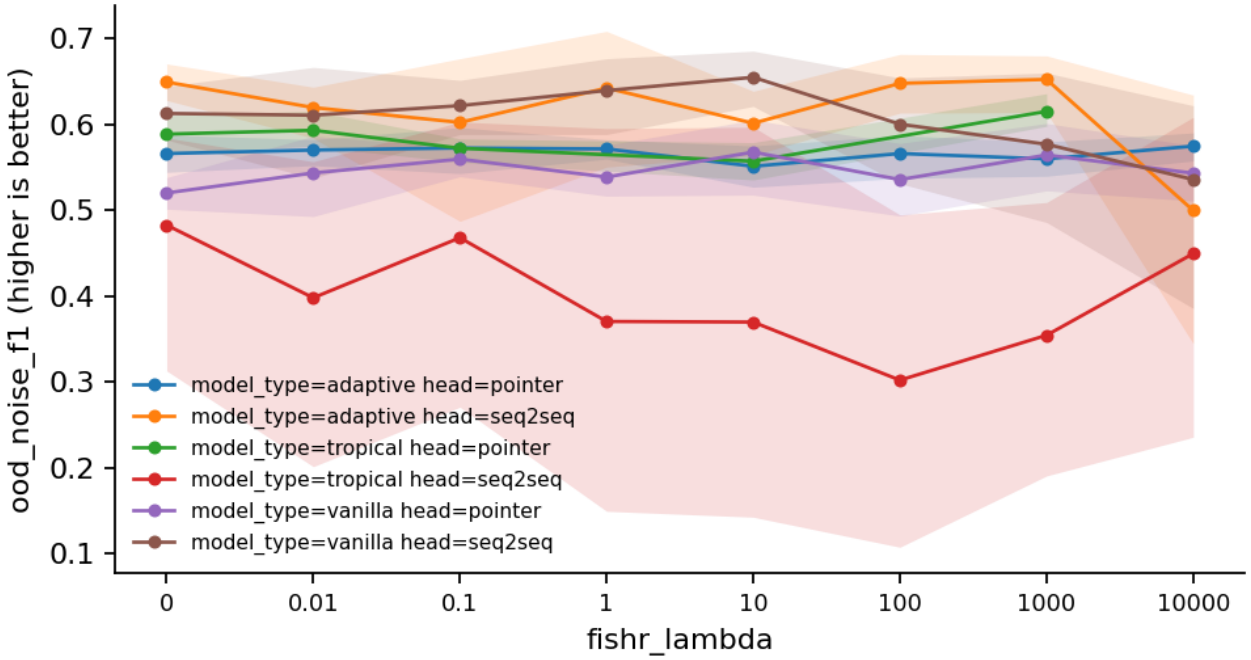


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

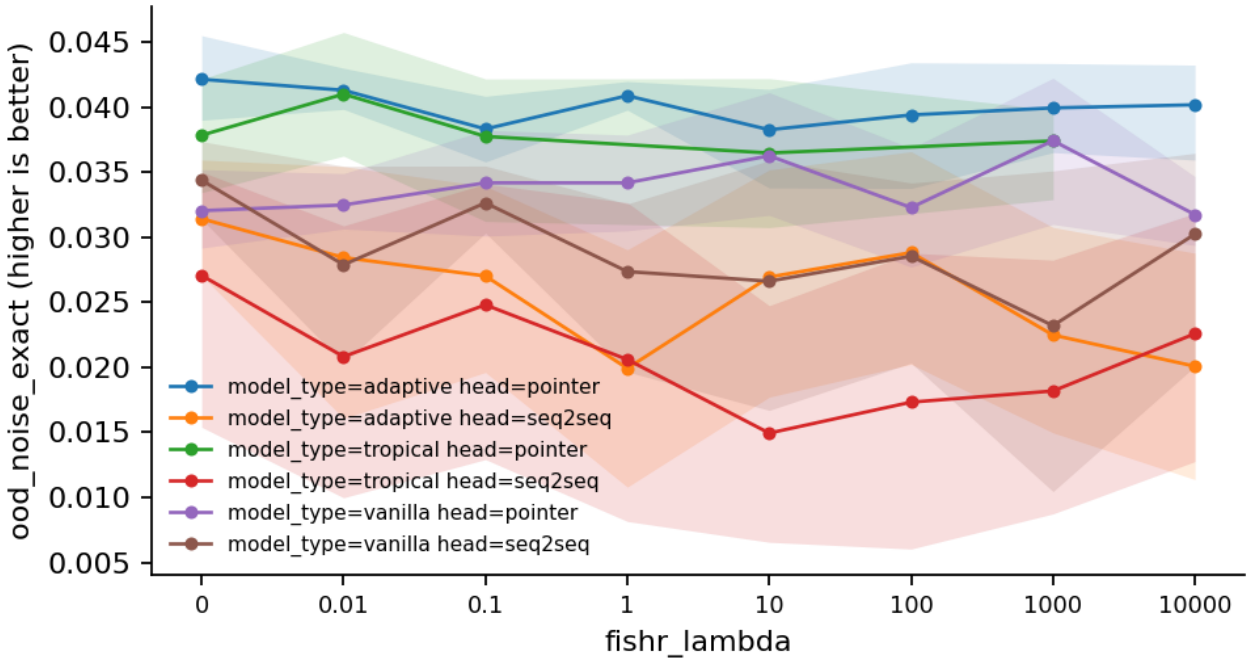


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-5.714×10^{-5}	$[-6.09 \times 10^{-4}, 4.14 \times 10^{-4}]$	0.8696	0.0876	0.5159
adaptive	seq2seq	-9.021×10^{-4}	$[-0.00262, 8.34 \times 10^{-4}]$	0.3342	-0.122	0.3719
tropical	pointer	-6.814×10^{-4}	$[-0.0025, 8.28 \times 10^{-4}]$	0.4125	-0.19	0.4457
tropical	seq2seq	-4.014×10^{-4}	$[-0.00165, 8.49 \times 10^{-4}]$	0.7167	-0.0381	0.7746
vanilla	pointer	8.429×10^{-5}	$[-4.51 \times 10^{-4}, 5.41 \times 10^{-4}]$	0.8514	0.0305	0.8318
vanilla	seq2seq	-3.793×10^{-4}	[-0.00197, 0.00101]	0.6717	0.0495	0.7199

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0058	[-0.00442, 0.016]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.008575	[-0.00442, 0.0216]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.001925	[-0.0135, 0.0173]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.002225	[-0.0109, 0.0153]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.0031	[-0.0211, 0.0273]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.00732	[-0.0231, 0.00844]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.00706	[-0.028, 0.0139]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.00784	[-0.0282, 0.0125]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.00676	[-0.0323, 0.0188]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.01168	[-0.0343, 0.011]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.01122	[-0.031, 0.0086]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.00502	[-0.0289, 0.0188]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.00766	[-0.0164, 0.00106]	0.9278
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.0101	[0.00483, 0.0154]	0.0957
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.00882	[0.00389, 0.0138]	0.1153
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.00414	[-0.00634, 0.0146]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.0067	$[8.85 \times 10^{-4}, 0.0125]$	0.4282
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.002	[-0.0103, 0.0143]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.00712	[0.00189, 0.0123]	0.2717
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.00252	[-0.00822, 0.0133]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.00844	[0.00111, 0.0158]	0.4282

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.00296	[-0.0131, 0.00714]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	5.600×10^{-4}	[-0.0119, 0.0131]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.00562	[-0.0201, 0.00882]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.00744	[-0.0232, 0.00828]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	3.200×10^{-4}	[-0.0271, 0.0277]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	2.800×10^{-4}	[-0.00982, 0.0104]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-7.200×10^{-4}	[-0.0148, 0.0134]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.01016	[-0.0355, 0.0151]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.01072	[0.00471, 0.0167]	0.1550
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.01286	[-0.00285, 0.0286]	0.9516
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.0113	[-0.00207, 0.0247]	0.9516
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.02096	[0.00799, 0.0339]	0.1859
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.01132	[-0.00775, 0.0304]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.01058	[0.00205, 0.0191]	0.3932
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.01746	[0.00883, 0.0261]	0.1038
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.0201	[0.00794, 0.0323]	0.1851
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.01078	[-0.00162, 0.0232]	0.9516
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.02413	[0.0111, 0.0371]	0.1851
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.01573	[0.00285, 0.0286]	0.4222
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.01995	[0.00686, 0.033]	0.2676
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.01593	[-0.0262, 0.0581]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.00234	[-0.0123, 0.00758]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.0046	[-0.00992, 0.0191]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.00154	[-0.00514, 0.00822]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.00682	[-0.00811, 0.0217]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.00964	[-0.00259, 0.0219]	0.9516
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.00374	[-0.00547, 0.013]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.01422	[-0.00891, 0.0374]	1.0000

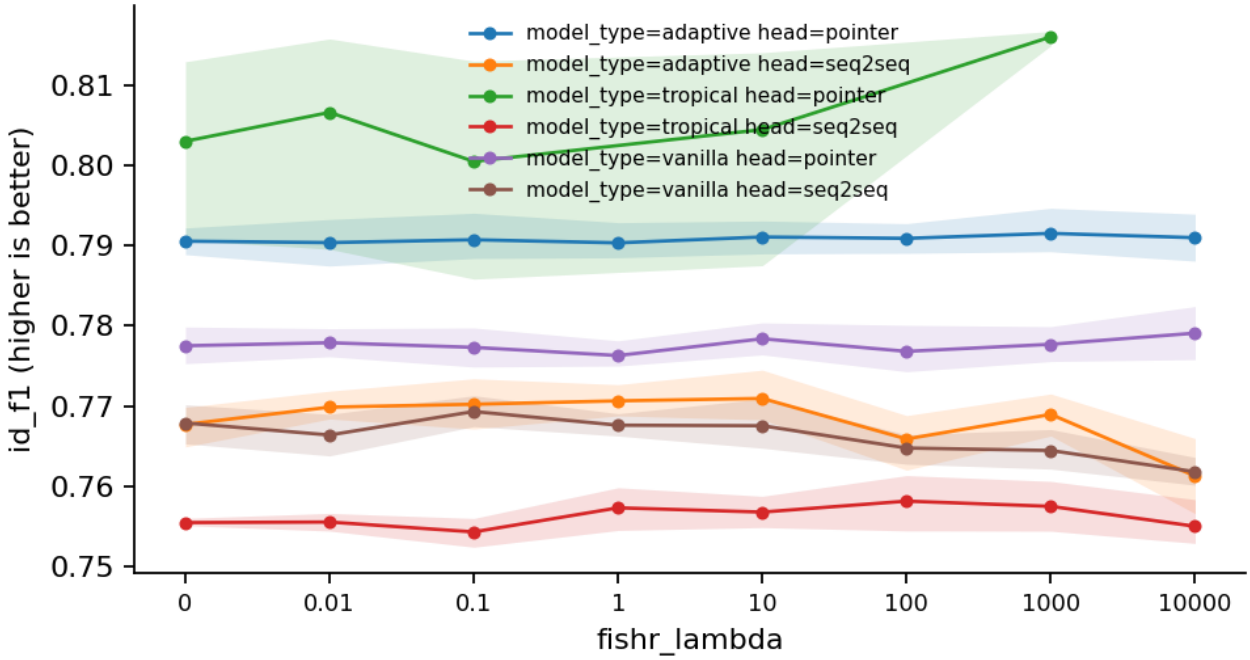


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.0015	[-0.0118, 0.0148]	1.0000

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.7906	[0.789, 0.792]	0.00221	5	0
adaptive	pointer	0.01	0.7904	[0.787, 0.793]	0.00383	5	0
adaptive	pointer	0.1	0.7907	[0.788, 0.794]	0.00379	5	0
adaptive	pointer	1	0.7904	[0.789, 0.793]	0.00283	5	0
adaptive	pointer	10	0.7911	[0.789, 0.793]	0.00252	5	0
adaptive	pointer	100	0.7909	[0.789, 0.793]	0.00242	5	0
adaptive	pointer	1000	0.7915	[0.789, 0.795]	0.00363	5	0
adaptive	pointer	1e+04	0.791	[0.788, 0.794]	0.00398	5	0
adaptive	seq2seq	0	0.7677	[0.765, 0.77]	0.00323	5	0
adaptive	seq2seq	0.01	0.7699	[0.768, 0.772]	0.00234	5	0
adaptive	seq2seq	0.1	0.7702	[0.767, 0.773]	0.00399	5	0
adaptive	seq2seq	1	0.7706	[0.769, 0.773]	0.00261	5	0
adaptive	seq2seq	10	0.7709	[0.768, 0.774]	0.00403	5	0
adaptive	seq2seq	100	0.7659	[0.762, 0.769]	0.00438	5	0
adaptive	seq2seq	1000	0.7689	[0.766, 0.772]	0.0036	5	0
adaptive	seq2seq	1e+04	0.7613	[0.757, 0.766]	0.00593	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0	0.803	[0.791, 0.813]	0.0148	5	0
tropical	pointer	0.01	0.8066	[0.79, 0.816]	0.017	4	0
tropical	pointer	0.1	0.8005	[0.786, 0.813]	0.0171	4	0
tropical	pointer	10	0.8045	[0.788, 0.814]	0.0168	4	0
tropical	pointer	1000	0.8161	[0.815, 0.817]	9.54×10^{-4}	3	0
tropical	seq2seq	0	0.7554	[0.755, 0.756]	6.04×10^{-4}	5	0
tropical	seq2seq	0.01	0.7555	[0.754, 0.757]	0.00147	5	0
tropical	seq2seq	0.1	0.7543	[0.752, 0.756]	0.00228	5	0
tropical	seq2seq	1	0.7573	[0.754, 0.76]	0.00339	5	0
tropical	seq2seq	10	0.7568	[0.755, 0.759]	0.00263	5	0
tropical	seq2seq	100	0.7581	[0.754, 0.761]	0.00432	5	0
tropical	seq2seq	1000	0.7575	[0.754, 0.761]	0.0041	5	0
tropical	seq2seq	1e+04	0.755	[0.753, 0.758]	0.00365	5	0
vanilla	pointer	0	0.7775	[0.775, 0.78]	0.00304	5	0
vanilla	pointer	0.01	0.7779	[0.776, 0.78]	0.00242	5	0
vanilla	pointer	0.1	0.7773	[0.775, 0.78]	0.00329	5	0
vanilla	pointer	1	0.7763	[0.775, 0.778]	0.0021	5	0
vanilla	pointer	10	0.7784	[0.776, 0.78]	0.00268	5	0
vanilla	pointer	100	0.7768	[0.774, 0.78]	0.00371	5	0
vanilla	pointer	1000	0.7777	[0.776, 0.78]	0.00291	5	0
vanilla	pointer	1e+04	0.7791	[0.776, 0.782]	0.00441	5	0
vanilla	seq2seq	0	0.7679	[0.765, 0.77]	0.00313	5	0
vanilla	seq2seq	0.01	0.7664	[0.764, 0.769]	0.00336	5	0
vanilla	seq2seq	0.1	0.7693	[0.767, 0.771]	0.00255	5	0
vanilla	seq2seq	1	0.7676	[0.766, 0.769]	0.00182	5	0
vanilla	seq2seq	10	0.7675	[0.765, 0.771]	0.00407	5	0
vanilla	seq2seq	100	0.7647	[0.763, 0.766]	0.00234	5	0
vanilla	seq2seq	1000	0.7644	[0.762, 0.767]	0.00329	5	0
vanilla	seq2seq	1e+04	0.7618	[0.76, 0.764]	0.00233	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-1.796×10^{-4}	[-0.00581, 0.00545]	1.0000
adaptive	pointer	0.1	1.776×10^{-4}	[-0.00545, 0.00581]	1.0000
adaptive	pointer	1	-2.097×10^{-4}	[-0.00584, 0.00542]	1.0000
adaptive	pointer	10	5.255×10^{-4}	[-0.00511, 0.00616]	0.9999
adaptive	pointer	100	3.523×10^{-4}	[-0.00528, 0.00598]	1.0000
adaptive	pointer	1000	9.769×10^{-4}	[-0.00465, 0.00661]	0.9964
adaptive	pointer	1e+04	4.365×10^{-4}	[-0.00519, 0.00607]	1.0000
adaptive	seq2seq	0.01	0.00216	[-0.00467, 0.00899]	0.9159
adaptive	seq2seq	0.1	0.002521	[-0.00431, 0.00935]	0.8435
adaptive	seq2seq	1	0.002943	[-0.00389, 0.00977]	0.7353
adaptive	seq2seq	10	0.003245	[-0.00358, 0.0101]	0.6497
adaptive	seq2seq	100	-0.001813	[-0.00864, 0.00502]	0.9626
adaptive	seq2seq	1000	0.001252	[-0.00558, 0.00808]	0.9950
adaptive	seq2seq	1e+04	-0.006432	[-0.0133, 3.97×10^{-4}]	0.0716
tropical	pointer	0.01	0.003604	[-0.0244, 0.0316]	0.9905
tropical	pointer	0.1	-0.00252	[-0.0306, 0.0255]	0.9976
tropical	pointer	10	0.001453	[-0.0266, 0.0295]	0.9997
tropical	pointer	1000	0.01303	[-0.0175, 0.0436]	0.6210
tropical	seq2seq	0.01	6.986×10^{-5}	[-0.00527, 0.00541]	1.0000
tropical	seq2seq	0.1	-0.001167	[-0.00651, 0.00418]	0.9865

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
tropical	seq2seq	1	0.001844	[-0.0035, 0.00719]	0.8790	
tropical	seq2seq	10	0.001311	[-0.00403, 0.00665]	0.9748	
tropical	seq2seq	100	0.002681	[-0.00266, 0.00802]	0.5977	
tropical	seq2seq	1000	0.002027	[-0.00332, 0.00737]	0.8269	
tropical	seq2seq	1e+04	-4.617×10^{-4}	[-0.0058, 0.00488]	1.0000	
vanilla	pointer	0.01	3.744×10^{-4}	[-0.00512, 0.00587]	1.0000	
vanilla	pointer	0.1	-2.022×10^{-4}	[-0.0057, 0.00529]	1.0000	
vanilla	pointer	1	-0.001222	[-0.00671, 0.00427]	0.9851	
vanilla	pointer	10	8.717×10^{-4}	[-0.00462, 0.00636]	0.9979	
vanilla	pointer	100	-7.031×10^{-4}	[-0.0062, 0.00479]	0.9995	
vanilla	pointer	1000	1.817×10^{-4}	[-0.00531, 0.00567]	1.0000	
vanilla	pointer	1e+04	0.001573	[-0.00392, 0.00707]	0.9463	
vanilla	seq2seq	0.01	-0.001499	[-0.00664, 0.00364]	0.9413	
vanilla	seq2seq	0.1	0.001436	[-0.0037, 0.00657]	0.9520	
vanilla	seq2seq	1	-2.732×10^{-4}	[-0.00541, 0.00486]	1.0000	
vanilla	seq2seq	10	-3.401×10^{-4}	[-0.00548, 0.0048]	1.0000	
vanilla	seq2seq	100	-0.003122	[-0.00826, 0.00201]	0.4026	
vanilla	seq2seq	1000	-0.003432	[-0.00857, 0.0017]	0.3096	
vanilla	seq2seq	1e+04	-0.006088	[-0.0112, -9.51×10^{-4}]	0.0147	*

Table 27: `id_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	1.432×10^{-4}	[5.39×10^{-5}, 2.16×10^{-4}]	0.5874	0.101	0.4577
adaptive	seq2seq	-0.001181	[-0.00157, -8.40×10^{-4}]	0.0029	-0.341	0.0094
tropical	pointer	0.001984	[4.59×10^{-5}, 0.00486]	0.3540	0.238	0.3293
tropical	seq2seq	2.011×10^{-4}	[-1.98×10^{-4}, 7.55×10^{-4}]	0.4726	0.101	0.4493
vanilla	pointer	1.743×10^{-4}	[-4.26×10^{-6}, 3.53×10^{-4}]	0.5000	0.0476	0.7445
vanilla	seq2seq	-9.412×10^{-4}	[-0.00147, -4.13×10^{-4}]	0.0008	-0.455	0.0010

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.02553	[0.00515, 0.0459]	0.1778	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.02819	[-0.00199, 0.0584]	0.2114	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.02291	[-0.00765, 0.0535]	0.2114	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.02675	[-0.00267, 0.0562]	0.2114	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.04032	[0.0354, 0.0452]	0.0096	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.01243	[-0.0162, -0.00863]	0.0096	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.01086	[-0.0154, -0.00628]	0.0275	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.01503	[-0.0189, -0.0112]	0.0054	*

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.01031	[-0.0149, -0.0057]	0.0310	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.01078	[-0.0168, -0.00473]	0.0621	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.006629	[-0.0123, -9.92×10^{-4}]	0.1855	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.006973	[-0.0141, 1.38×10^{-4}]	0.2114	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.006806	[-0.0132, -4.61×10^{-4}]	0.2041	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.01305	[0.0101, 0.016]	0.0034	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.0125	[0.00903, 0.016]	0.0062	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.01343	[0.00855, 0.0183]	0.0158	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.01406	[0.0119, 0.0163]	0.0009	*
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.01271	[0.00927, 0.0161]	0.0061	*
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.01411	[0.0104, 0.0178]	0.0057	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.01385	[0.0118, 0.0159]	0.0007	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.01192	[0.00515, 0.0187]	0.0722	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-1.788×10^{-4}	[-0.00704, 0.00668]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.00348	[-0.00278, 0.00974]	0.9885	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	9.065×10^{-4}	[-0.00336, 0.00517]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.003038	[0.00132, 0.00476]	0.0722	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.003407	[6.83×10^{-4}, 0.00613]	0.1787	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.001131	[-0.006, 0.00827]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.004505	[8.22×10^{-4}, 0.00819]	0.1787	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-5.233×10^{-4}	[-0.00835, 0.00731]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.02288	[0.0168, 0.0289]	0.0064	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.02054	[0.0175, 0.0236]	0.0009	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.02053	[0.0159, 0.0251]	0.0039	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.01972	[0.0187, 0.0208]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.02016	[0.0178, 0.0225]	0.0003	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.02504	[0.0171, 0.0329]	0.0120	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.0226	[0.0194, 0.0258]	0.0008	*

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.02975	[0.0223, 0.0372]	0.0057	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.0476	[0.0285, 0.0667]	0.0229	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.05063	[0.0242, 0.0771]	0.0443	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.04548	[0.0168, 0.0742]	0.0443	*
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.04853	[0.0207, 0.0763]	0.0443	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.06124	[0.0525, 0.07]	0.0132	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.009645	[0.00529, 0.014]	0.0304	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.01152	[0.0054, 0.0176]	0.0383	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.008007	[0.00236, 0.0137]	0.0443	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.008697	[0.00679, 0.0106]	0.0038	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.01086	[0.00453, 0.0172]	0.0443	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.01206	[0.00669, 0.0174]	0.0304	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.01326	[0.00701, 0.0195]	0.0304	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.01731	[0.0109, 0.0237]	0.0185	*

A.7 `id_exact`

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.1967	[0.192, 0.202]	0.00674	5	0
adaptive	pointer	0.01	0.1917	[0.185, 0.202]	0.011	5	0
adaptive	pointer	0.1	0.1922	[0.184, 0.205]	0.0139	5	0
adaptive	pointer	1	0.1912	[0.185, 0.199]	0.00922	5	0
adaptive	pointer	10	0.1929	[0.187, 0.2]	0.00863	5	0
adaptive	pointer	100	0.1949	[0.187, 0.203]	0.01	5	0
adaptive	pointer	1000	0.1969	[0.19, 0.207]	0.0113	5	0
adaptive	pointer	1e+04	0.1919	[0.182, 0.203]	0.014	5	0
adaptive	seq2seq	0	0.1079	[0.0938, 0.12]	0.0171	5	0
adaptive	seq2seq	0.01	0.1198	[0.114, 0.125]	0.00729	5	0
adaptive	seq2seq	0.1	0.1199	[0.113, 0.125]	0.00826	5	0
adaptive	seq2seq	1	0.117	[0.109, 0.125]	0.0116	5	0
adaptive	seq2seq	10	0.1151	[0.106, 0.124]	0.012	5	0
adaptive	seq2seq	100	0.1039	[0.0927, 0.115]	0.014	5	0
adaptive	seq2seq	1000	0.111	[0.103, 0.119]	0.0104	5	0
adaptive	seq2seq	1e+04	0.09386	[0.0817, 0.103]	0.014	5	0
tropical	pointer	0	0.2544	[0.205, 0.292]	0.0577	5	0
tropical	pointer	0.01	0.2667	[0.21, 0.298]	0.0568	4	0
tropical	pointer	0.1	0.2472	[0.195, 0.286]	0.0572	4	0
tropical	pointer	10	0.2566	[0.197, 0.289]	0.0596	4	0

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	0.285	[0.279, 0.29]	0.00551	3	0
tropical	seq2seq	0	0.08428	[0.0817, 0.0868]	0.00326	5	0
tropical	seq2seq	0.01	0.0875	[0.0826, 0.092]	0.0059	5	0
tropical	seq2seq	0.1	0.08462	[0.0816, 0.0869]	0.00335	5	0
tropical	seq2seq	1	0.08504	[0.0811, 0.0889]	0.00513	5	0
tropical	seq2seq	10	0.08614	[0.0836, 0.09]	0.00447	5	0
tropical	seq2seq	100	0.08818	[0.0857, 0.0905]	0.00329	5	0
tropical	seq2seq	1000	0.0827	[0.0809, 0.0846]	0.00238	5	0
tropical	seq2seq	1e+04	0.08186	[0.0774, 0.0866]	0.0062	5	0
vanilla	pointer	0	0.1905	[0.174, 0.207]	0.0214	5	0
vanilla	pointer	0.01	0.1854	[0.168, 0.203]	0.0211	5	0
vanilla	pointer	0.1	0.1859	[0.171, 0.199]	0.0183	5	0
vanilla	pointer	1	0.1833	[0.17, 0.197]	0.0176	5	0
vanilla	pointer	10	0.1864	[0.168, 0.203]	0.0223	5	0
vanilla	pointer	100	0.1867	[0.17, 0.206]	0.0226	5	0
vanilla	pointer	1000	0.1867	[0.17, 0.204]	0.0224	5	0
vanilla	pointer	1e+04	0.1887	[0.175, 0.205]	0.0191	5	0
vanilla	seq2seq	0	0.09694	[0.0921, 0.102]	0.00644	5	0
vanilla	seq2seq	0.01	0.0984	[0.0919, 0.105]	0.00875	5	0
vanilla	seq2seq	0.1	0.1014	[0.0955, 0.107]	0.00734	5	0
vanilla	seq2seq	1	0.1	[0.0941, 0.107]	0.00793	5	0
vanilla	seq2seq	10	0.1	[0.0927, 0.107]	0.00961	5	0
vanilla	seq2seq	100	0.09558	[0.0819, 0.104]	0.0147	5	0
vanilla	seq2seq	1000	0.08764	[0.0794, 0.0933]	0.00951	5	0
vanilla	seq2seq	1e+04	0.07986	[0.0722, 0.0876]	0.0102	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.00498	[-0.0239, 0.014]	0.9645
adaptive	pointer	0.1	-0.00448	[-0.0234, 0.0145]	0.9794
adaptive	pointer	1	-0.00546	[-0.0244, 0.0135]	0.9448
adaptive	pointer	10	-0.0038	[-0.0228, 0.0152]	0.9916
adaptive	pointer	100	-0.00184	[-0.0208, 0.0171]	0.9999
adaptive	pointer	1000	1.600×10^{-4}	[-0.0188, 0.0191]	1.0000
adaptive	pointer	1e+04	-0.00484	[-0.0238, 0.0141]	0.9692
adaptive	seq2seq	0.01	0.01194	[-0.00938, 0.0333]	0.4868
adaptive	seq2seq	0.1	0.01198	[-0.00934, 0.0333]	0.4834
adaptive	seq2seq	1	0.00906	[-0.0123, 0.0304]	0.7468
adaptive	seq2seq	10	0.00718	[-0.0141, 0.0285]	0.8906
adaptive	seq2seq	100	-0.00396	[-0.0253, 0.0174]	0.9946
adaptive	seq2seq	1000	0.00306	[-0.0183, 0.0244]	0.9989
adaptive	seq2seq	1e+04	-0.01404	[-0.0354, 0.00728]	0.3235
tropical	pointer	0.01	0.01227	[-0.0871, 0.112]	0.9918
tropical	pointer	0.1	-0.007175	[-0.107, 0.0922]	0.9990
tropical	pointer	10	0.00215	[-0.0972, 0.102]	1.0000
tropical	pointer	1000	0.0306	[-0.0776, 0.139]	0.8639
tropical	seq2seq	0.01	0.00322	[-0.00454, 0.011]	0.7649
tropical	seq2seq	0.1	3.400×10^{-4}	[-0.00742, 0.0081]	1.0000
tropical	seq2seq	1	7.600×10^{-4}	[-0.007, 0.00852]	0.9999
tropical	seq2seq	10	0.00186	[-0.0059, 0.00962]	0.9777
tropical	seq2seq	100	0.0039	[-0.00386, 0.0117]	0.5959
tropical	seq2seq	1000	-0.00158	[-0.00934, 0.00618]	0.9908

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
tropical	seq2seq	1e+04	-0.00242	[-0.0102, 0.00534]	0.9208	
vanilla	pointer	0.01	-0.0051	[-0.0412, 0.031]	0.9990	
vanilla	pointer	0.1	-0.00466	[-0.0408, 0.0315]	0.9994	
vanilla	pointer	1	-0.0072	[-0.0433, 0.0289]	0.9919	
vanilla	pointer	10	-0.00418	[-0.0403, 0.0319]	0.9997	
vanilla	pointer	100	-0.00386	[-0.04, 0.0323]	0.9998	
vanilla	pointer	1000	-0.0038	[-0.0399, 0.0323]	0.9998	
vanilla	pointer	1e+04	-0.00184	[-0.038, 0.0343]	1.0000	
vanilla	seq2seq	0.01	0.00146	[-0.0153, 0.0182]	1.0000	
vanilla	seq2seq	0.1	0.00444	[-0.0123, 0.0212]	0.9631	
vanilla	seq2seq	1	0.00306	[-0.0137, 0.0198]	0.9951	
vanilla	seq2seq	10	0.0031	[-0.0137, 0.0199]	0.9948	
vanilla	seq2seq	100	-0.00136	[-0.0181, 0.0154]	1.0000	
vanilla	seq2seq	1000	-0.0093	[-0.0261, 0.00747]	0.4969	
vanilla	seq2seq	1e+04	-0.01708	[-0.0339, -3.08×10^{-4}]	0.0446	*

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	4.757×10^{-4}	[3.04×10^{-4}, 6.88×10^{-4}]	0.5977	0.135	0.3053
adaptive	seq2seq	-0.003886	[-0.00518, -0.00294]	0.0003	-0.446	0.0004
tropical	pointer	0.004053	[-0.0018, 0.0142]	0.5702	-0.167	0.5013
tropical	seq2seq	-6.293×10^{-4}	[-0.00153, 2.61×10^{-4}]	0.1148	-0.217	0.1052
vanilla	pointer	5.300×10^{-4}	[-7.33×10^{-4}, 0.00186]	0.7431	0.0457	0.7690
vanilla	seq2seq	-0.003126	[-0.00569, -9.45×10^{-4}]	0.0008	-0.429	0.0004

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.06386	[-0.0326, 0.16]	0.8861	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.0812	[-0.0419, 0.204]	0.8861	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.05835	[-0.0574, 0.174]	0.8861	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.07135	[-0.0556, 0.198]	0.8861	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.1122	[0.0884, 0.136]	0.0315	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.01266	[-0.0228, -0.00253]	0.2559	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.0109	[-0.0249, 0.00314]	0.7784	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.01676	[-0.0289, -0.00459]	0.2061	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.01496	[-0.0251, -0.00478]	0.1813	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.0139	[-0.0293, 0.00149]	0.5965	

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.0074	[-0.0264, 0.0116]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.00494	[-0.0188, 0.00889]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.002	[-0.0177, 0.0217]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00616	[-0.0211, 0.0335]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.00628	[-0.012, 0.0246]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.00634	[-0.0104, 0.0231]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.0079	[-0.0115, 0.0273]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.00654	[-0.0179, 0.0309]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.00818	[-0.0156, 0.0319]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.01012	[-0.00689, 0.0271]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.00316	[-0.0275, 0.0338]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.01096	[-0.0136, 0.0355]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.02144	[0.00331, 0.0396]	0.4258	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.0185	[0.00258, 0.0344]	0.4258	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.01696	[0.00522, 0.0287]	0.2395	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01504	[-0.00355, 0.0336]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.00836	[-0.0191, 0.0358]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.02332	[0.0137, 0.033]	0.0408	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.014	[-0.00707, 0.0351]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.0888	[0.0684, 0.109]	0.0043	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.07188	[0.0546, 0.0891]	0.0048	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.07234	[0.0541, 0.0906]	0.0050	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.07428	[0.0583, 0.0903]	0.0038	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.07782	[0.0701, 0.0855]	0.0002	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.09092	[0.0617, 0.12]	0.0065	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.0859	[0.0737, 0.0981]	0.0008	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.098	[0.0743, 0.122]	0.0048	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.1701	[0.102, 0.239]	0.0092	*

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.1809	[0.0967, 0.265]	0.0192	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.1634	[0.0773, 0.249]	0.0192	*
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.1695	[0.0755, 0.264]	0.0192	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.2022	[0.192, 0.212]	0.0024	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.0936	[0.0674, 0.12]	0.0052	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.08704	[0.0621, 0.112]	0.0052	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.0845	[0.0654, 0.104]	0.0043	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.08334	[0.0618, 0.105]	0.0050	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.08632	[0.0642, 0.108]	0.0050	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.0911	[0.0613, 0.121]	0.0065	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.0991	[0.0678, 0.13]	0.0065	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.1088	[0.0808, 0.137]	0.0050	*

B Curves

C Configuration

The base configuration every run shares (frozen hash `beab4c1c6902`); factor levels are in Table 1.

```
task:
  name: BalancedPartitionDataset
  value_range:
  - 1
  - 10
  target_range: null
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - 11
  - 100
  ood_noise_range:
  - 10
  - 30
```

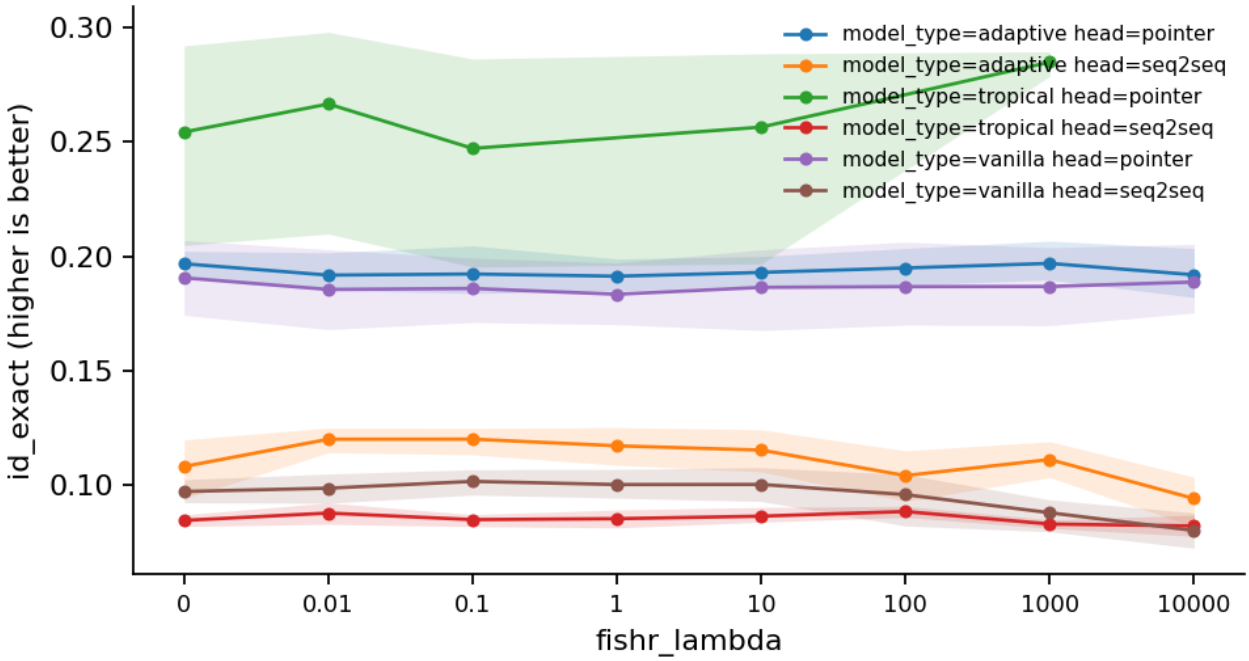


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

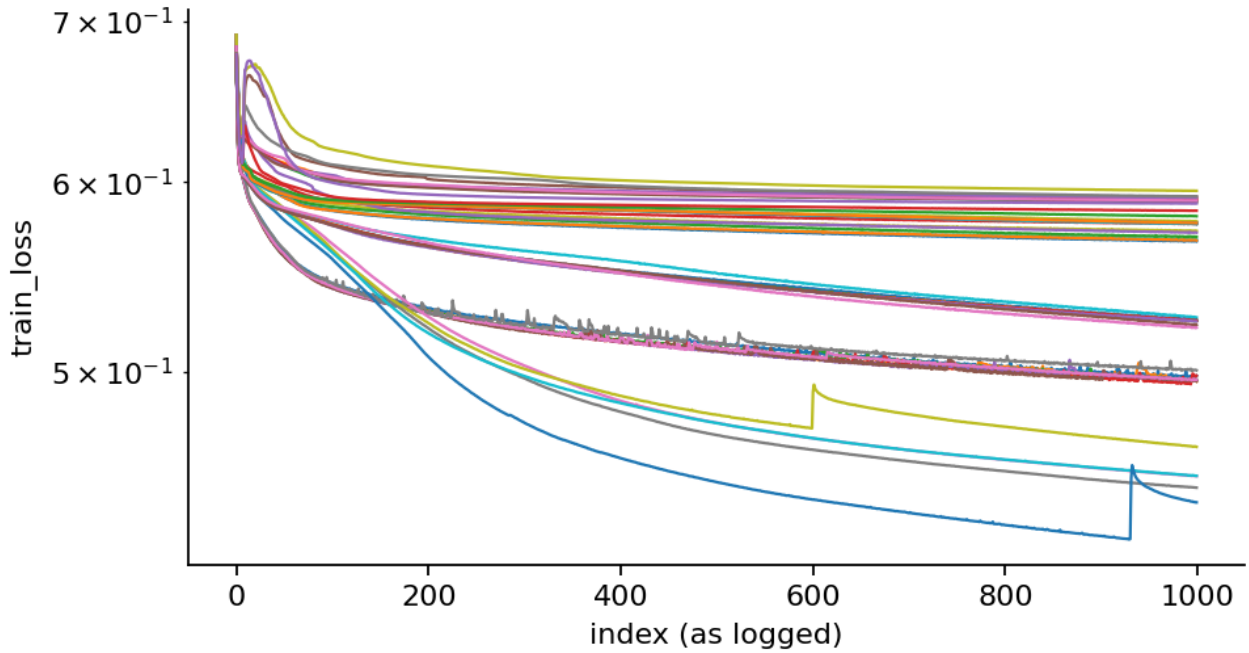


Figure 9: `train_loss`, the mean over seeds per cell.

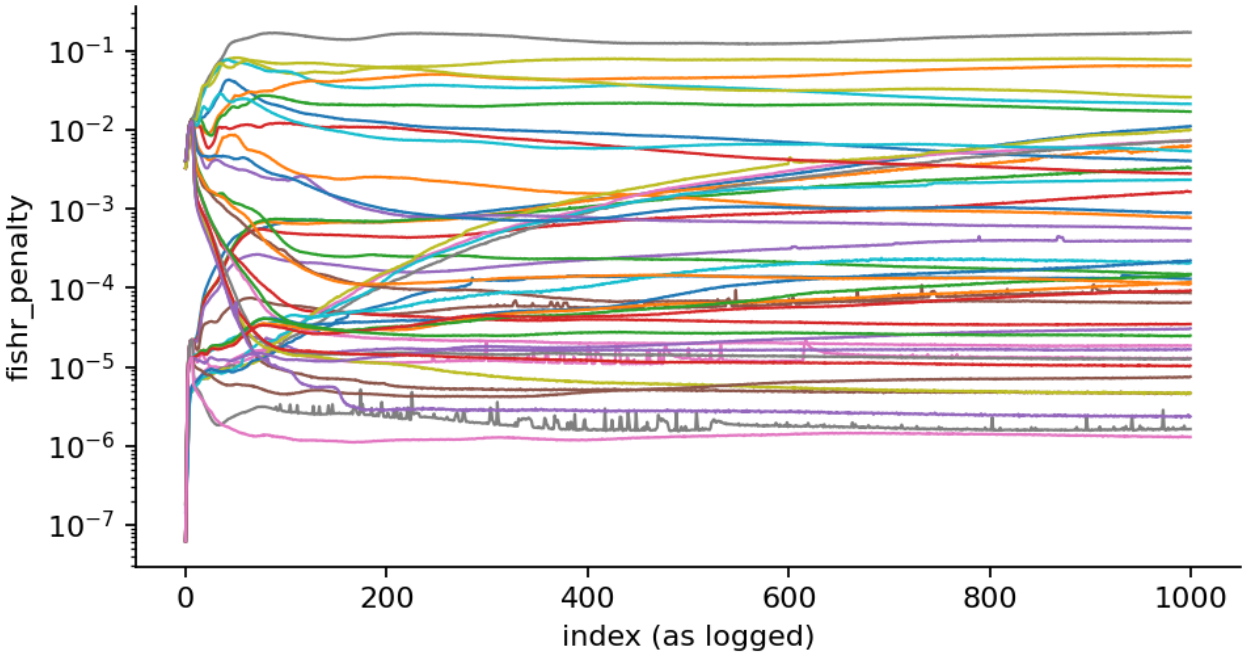


Figure 10: `fishr_penalty`, the mean over seeds per cell.

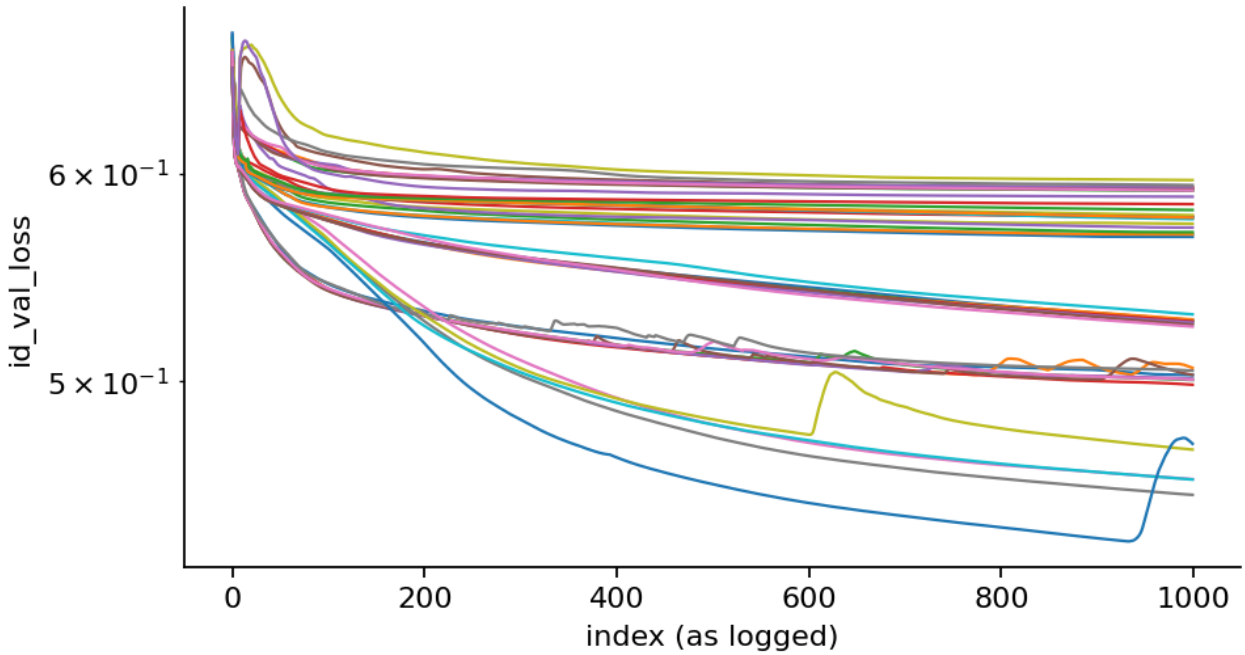


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 1000
  batch_size: 100
  eval_batch_size: 256
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/BalancedPartition/results`.